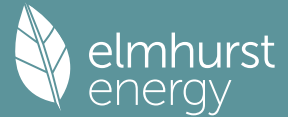


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Property Reference	CPG002989 - Planning				Issued on Date	05/06/2025
Assessment Reference	Land Adj to Piskey Laden	Prop Type Ref		Land Adj to Piskey Laden		
Property	Land Adj to Piskey Laden, Gwallon Lane, Marazion, Cornwall , TR17 0HJ					
SAP Rating	98 A	DER	-1.18	TER	10.49	
Environmental	101 A	% DER < TER				111.25
CO ₂ Emissions (t/year)	-0.18	DFEE	41.20	TFEE	48.36	
Compliance Check	See BREL	% DFEE < TFEE				14.80
% DPER < TPER	87.75	DPER	6.87	TPER	56.09	
Assessor Details	Mr. Aaron Roberts				Assessor ID	DV31-0001
Client	Mr P Else , Mr P Else					

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF DWELLING EMISSIONS FOR REGULATIONS COMPLIANCE

1. Overall dwelling characteristics

	Area (m ²)	Storey height (m)	Volume (m ³)
Main dwelling	70.1100 (1b)	2.6300 (2b)	184.3893 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100		184.3893 (4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n)	184.3893 (5)

2. Ventilation rate

	m ³ per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 = 0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	0 * 10 = 0.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)

Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	0.0000 / (5) =	0.0000 (8)
Pressure test	Yes	
Pressure Test Method	Blower Door	
Measured/design AP50	1.0000	(17)
Infiltration rate	0.0500	(18)
Number of sides sheltered	0	(19)

Shelter factor	(20) = 1 - [0.075 x (19)] =	1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) =	0.0500 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.0638	0.0625	0.0613	0.0550	0.0537	0.0475	0.0475	0.0463	0.0500	0.0537	0.0563	0.0588 (22b)
Balanced mechanical ventilation with heat recovery												
If mechanical ventilation												0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												84.6000 (23c)
Effective ac	0.1407	0.1395	0.1382	0.1320	0.1307	0.1245	0.1245	0.1232	0.1270	0.1307	0.1332	0.1357 (25)

3. Heat losses and heat loss parameter

Element	Gross m ²	Openings m ²	NetArea m ²	U-value W/m ² K	A x U W/K	K-value kJ/m ² K	A x K kJ/K
Main dwelling							
Door			2.1200	1.0000	2.1200		(26)
Window			20.5800	1.1450	23.5649		(27)
Heatloss Floor 1			70.1100	0.1200	8.4132	110.0000	7712.1000 (28a)
External Wall 1	81.6400	22.7000	58.9400	0.1600	9.4304	9.0000	530.4600 (29a)
External Roof 1	72.5800		72.5800	0.1400	10.1612	9.0000	653.2200 (30)
Total net area of external elements Aum(A, m ²)			224.3300				(31)
Fabric heat loss, W/K = Sum (A x U)				(26)...(30) + (32) =	53.6897		(33)
Main dwelling							
Internal Wall 1			127.1400			9.0000	1144.2600 (32c)
Heat capacity Cm = Sum(A x k)					(28)...(30) + (32) + (32a)...(32e) =		10040.0400 (34)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m ² K							143.2041 (35)
List of Thermal Bridges							
K1 Element				Length	Psi-value	Total	
E2 Other lintels (including other steel lintels)				13.0700	0.0840	1.0979	

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E4 Jamb						32.8000		0.0430		1.4104			
E5 Ground floor (normal)						35.9600		0.0210		0.7552			
E3 Sill						4.9700		0.0340		0.1690			
E11 Eaves (insulation at rafter level)						24.2600		0.0400		0.9704			
E13 Gable (insulation at rafter level)						11.8000		0.0240		0.2832			
E16 Corner (normal)						8.9200		0.0300		0.2676			
Thermal bridges (Sum(L x Psi) calculated using Appendix K)											4.9536	(36)	
Point Thermal bridges											0.0000		
Total fabric heat loss									(33) + (36) + (36a) =		58.6433	(37)	
Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)													
(38)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Heat transfer coeff	8.5644	8.4884	8.4123	8.0320	7.9559	7.5756	7.5756	7.4996	7.7278	7.9559	8.1081	8.2602	(38)
Average = Sum(39)m / 12 =	67.2077	67.1317	67.0556	66.6753	66.5992	66.2189	66.2189	66.1429	66.3711	66.5992	66.7514	66.9035	(39)
												66.6563	
HLP	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
HLP (average)	0.9586	0.9575	0.9564	0.9510	0.9499	0.9445	0.9445	0.9434	0.9467	0.9499	0.9521	0.9543	(40)
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31	

4. Water heating energy requirements (kWh/year)													

Assumed occupancy												2.2488	(42)
Hot water usage for mixer showers													
	79.5240	78.3289	76.5874	73.2554	70.7965	68.0543	66.4956	68.2239	70.1185	73.0627	76.4663	79.2193	(42a)
Hot water usage for baths													
	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(42b)
Hot water usage for other uses													
	37.6599	36.2904	34.9210	33.5515	32.1821	30.8126	30.8126	32.1821	33.5515	34.9210	36.2904	37.6599	(42c)
Average daily hot water use (litres/day)												107.5547	(43)
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
	117.1839	114.6193	111.5084	106.8070	102.9786	98.8669	97.3083	100.4060	103.6700	107.9837	112.7568	116.8792	(44)
Energy conte	185.5906	163.2164	171.3843	146.2134	138.6289	121.5684	117.6859	124.3236	127.8355	146.5329	160.6427	183.0046	(45)
Energy content (annual)													
Distribution loss (46)m = 0.15 x (45)m												1786.6272	
	27.8386	24.4825	25.7076	21.9320	20.7943	18.2353	17.6529	18.6485	19.1753	21.9799	24.0964	27.4507	(46)
Water storage loss:													
Store volume												150.0000	(47)
a) If manufacturer declared loss factor is known (kWh/day):												1.9900	(48)
Temperature factor from Table 2b												0.5400	(49)
Enter (49) or (54) in (55)												1.0746	(55)
Total storage loss													
	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126	(56)
If cylinder contains dedicated solar storage													
	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126	(57)
Primary loss	23.2624	21.0112	23.2624	22.5120	23.2624	22.5120	23.2624	23.2624	22.5120	23.2624	22.5120	23.2624	(59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(61)
Total heat required for water heating calculated for each month													
	242.1656	214.3164	227.9593	200.9634	195.2039	176.3184	174.2609	180.8986	182.5855	203.1079	215.3927	239.5796	(62)
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	(63b)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(63d)
Output from w/h													
	242.1656	214.3164	227.9593	200.9634	195.2039	176.3184	174.2609	180.8986	182.5855	203.1079	215.3927	239.5796	(64)
12Total per year (kWh/year)												2452.7522	(64)
Electric shower(s)												2453	(64)
	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(64a)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =												0.0000	(64a)
Heat gains from water heating, kWh/month													
	106.9689	95.1495	102.2453	92.4159	91.3541	84.2215	84.3906	86.5976	86.3053	93.9822	97.2137	106.1090	(65)

5. Internal gains (see Table 5 and 5a)													

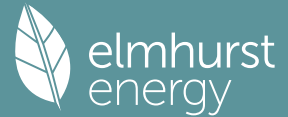
Metabolic gains (Table 5), Watts													
(66)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	(66)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	99.6431	110.3192	99.6431	102.9646	99.6431	102.9646	99.6431	99.6431	102.9646	99.6431	102.9646	99.6431	(67)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	197.5537	199.6037	194.4377	183.4401	169.5576	156.5101	147.7935	145.7435	150.9094	161.9070	175.7895	188.8371	(68)
Pumps, fans	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	(69)
Losses e.g. evaporation (negative values) (Table 5)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(70)
Water heating gains (Table 5)	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	(71)
	143.7754	141.5915	137.4264	128.3555	122.7878	116.9743	113.4282	116.3946	119.8685	126.3201	135.0190	142.6197	(72)
Total internal gains	497.7041	508.2462	488.2392	471.4921	448.7205	433.1808	417.5967	418.5131	430.4743	444.6022	470.5050	487.8317	(73)

6. Solar gains													

[Jan]													
	Area												
	m2												
	Solar flux												
	Table 6a												
	W/m2												
	Specific data												
	or Table 6b												
	Specific data												
	or Table 6c												
	Access												
	factor												
	Table 6d												
	Gains												
	W												
North	0.8500						0.7600		0.7000		0.7700	3.3322	(74)
East	1.8100						0.7600		0.7000		0.7700	13.1060	(76)
South	3.3600						0.7600		0.7000		0.7700	57.9142	(78)
West	14.5600						0.7600		0.7000		0.7700	105.4274	(80)

Solar gains	179.7799	333.0931	513.5087	710.8654	848.2508	860.7078	822.3937	719.8947	583.3490	385.0244	220.5561	150.2972	(83)
Total gains	677.4839	841.3393	1001.7479	1182.3574	1296.9712	1293.8886	1239.9904	1138.4078	1013.8234	829.6266	691.0610	638.1290	(84)

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7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	41.4967	41.5437	41.5909	41.8281	41.8759	42.1163	42.1163	42.1648	42.0198	41.8759	41.7804	41.6854
alpha	3.7664	3.7696	3.7727	3.7885	3.7917	3.8078	3.8078	3.8110	3.8013	3.7917	3.7854	3.7790
util living area	0.9349	0.8764	0.7785	0.6217	0.4620	0.3244	0.2342	0.2660	0.4394	0.7147	0.8893	0.9455 (86)
Living	20.0244	20.2902	20.5663	20.7872	20.8816	20.9115	20.9173	20.9164	20.8959	20.7360	20.3449	19.9656
Non living	18.9890	19.3134	19.6406	19.8913	19.9891	20.0208	20.0251	20.0255	20.0071	19.8455	19.3904	18.9192
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.5009	20.2902	20.5663	20.7872	20.8816	20.9115	20.9173	20.9164	20.8959	20.7360	20.3449	20.1103 (87)
Th 2	20.1180	20.1189	20.1198	20.1244	20.1253	20.1298	20.1298	20.1308	20.1280	20.1253	20.1234	20.1216 (88)
util rest of house	0.9247	0.8595	0.7525	0.5867	0.4222	0.2813	0.1882	0.2163	0.3876	0.6747	0.8709	0.9367 (89)
MIT 2	19.6679	19.3134	19.6406	19.8913	19.9891	20.0208	20.0251	20.0255	20.0071	19.8455	19.3904	19.1358 (90)
Living area fraction									fLA =	Living area / (4) =		0.3487 (91)
MIT	19.9584	19.6541	19.9635	20.2038	20.3003	20.3314	20.3363	20.3362	20.3170	20.1561	19.7233	19.4757 (92)
Temperature adjustment												0.0000
adjusted MIT	19.9584	19.6541	19.9635	20.2038	20.3003	20.3314	20.3363	20.3362	20.3170	20.1561	19.7233	19.4757 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Utilisation	0.9227	0.8495	0.7475	0.5900	0.4303	0.2914	0.1992	0.2281	0.3990	0.6762	0.8614	0.9292 (94)
Useful gains	625.1230	714.7216	748.8532	697.5628	558.0601	377.0087	246.9830	259.6253	404.4947	561.0161	595.3086	592.9428 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	14.6000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	1052.3658	990.4653	902.7999	753.6813	572.7762	379.5290	247.4111	260.3519	412.6307	636.4264	842.6218	1021.9958 (97)
Space heating kWh	317.8686	185.2998	114.5363	40.4053	10.9488	0.0000	0.0000	0.0000	0.0000	56.1052	178.0655	319.2154 (98a)
Space heating requirement - total per year (kWh/year)												1222.4450
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	317.8686	185.2998	114.5363	40.4053	10.9488	0.0000	0.0000	0.0000	0.0000	56.1052	178.0655	319.2154 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												1222.4450
Space heating per m2										(98c) / (4) =		17.4361 (99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)
Fraction of space heat from main system(s)												1.0000 (202)
Efficiency of main space heating system 1 (in %)												401.8421 (206)
Efficiency of main space heating system 2 (in %)												0.0000 (207)
Efficiency of secondary/supplementary heating system, %												0.0000 (208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Space heating requirement	317.8686	185.2998	114.5363	40.4053	10.9488	0.0000	0.0000	0.0000	0.0000	56.1052	178.0655	319.2154 (98)
Space heating efficiency (main heating system 1)	401.8421	401.8421	401.8421	401.8421	401.8421	0.0000	0.0000	0.0000	0.0000	401.8421	401.8421	401.8421 (210)
Space heating fuel (main heating system)	79.1029	46.1126	28.5028	10.0550	2.7246	0.0000	0.0000	0.0000	0.0000	13.9620	44.3123	79.4380 (211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating												
Water heating requirement	242.1656	214.3164	227.9593	200.9634	195.2039	176.3184	174.2609	180.8986	182.5855	203.1079	215.3927	239.5796 (64)
Efficiency of water heater (217)m	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890 (216)
Fuel for water heating, kWh/month	118.3669	104.7546	111.4230	98.2278	95.4127	86.1818	85.1761	88.4205	89.2450	99.2760	105.2807	117.1028 (219)
Space cooling fuel requirement												
(221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (221)
Pumps and Fa	11.5590	10.4404	11.5590	11.1861	11.5590	11.1861	11.5590	11.5590	11.1861	11.5590	11.1861	11.5590 (231)
Lighting	22.9836	18.4383	16.6017	12.1631	9.3951	7.6759	8.5705	11.1403	14.4702	18.9857	21.4443	23.6224 (232)
Electricity generated by PVs (Appendix M) (negative quantity)												
(233a)m	-39.4149	-62.1055	-99.1050	-117.4621	-129.0076	-120.7662	-118.8555	-111.0796	-94.6886	-73.4244	-44.8078	-32.8817 (233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)												
(234a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												
(235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)												
(235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235c)
Electricity generated by PVs (Appendix M) (negative quantity)												
(233b)m	-16.2958	-42.4039	-106.5961	-193.5779	-282.3947	-293.9538	-286.1186	-225.3275	-143.7754	-67.9951	-23.6210	-12.1154 (233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)												
(234b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												
(235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)												
(235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235d)
Annual totals kWh/year												
Space heating fuel - main system 1												304.2103 (211)
Space heating fuel - main system 2												0.0000 (213)
Space heating fuel - secondary												0.0000 (215)
Efficiency of water heater												204.5890
Water heating fuel used												1198.8679 (219)
Space cooling fuel												0.0000 (221)
Electricity for pumps and fans:												

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(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6050)	
mechanical ventilation fans (SFP = 0.6050)	136.0977 (230a)
Total electricity for the above, kWh/year	136.0977 (231)
Electricity for lighting (calculated in Appendix L)	185.4911 (232)
Energy saving/generation technologies (Appendices M ,N and Q)	
PV generation	-2737.7741 (233)
Wind generation	0.0000 (234)
Hydro-electric generation (Appendix N)	0.0000 (235a)
Electricity generated - Micro CHP (Appendix N)	0.0000 (235)
Appendix Q - special features	
Energy saved or generated	-0.0000 (236)
Energy used	0.0000 (237)
Total delivered energy for all uses	-913.1070 (238)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year
Space heating - main system 1	304.2103	0.1577	47.9686 (261)
Total CO2 associated with community systems			0.0000 (373)
Water heating (other fuel)	1198.8679	0.1408	168.8173 (264)
Space and water heating			216.7860 (265)
Pumps, fans and electric keep-hot	136.0977	0.1387	18.8784 (267)
Energy for lighting	185.4911	0.1443	26.7721 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1043.5990	0.1333	-139.1399
PV Unit electricity exported	-1694.1751	0.1216	-205.9661
Total			-345.1060 (269)
Total CO2, kg/year			-82.6695 (272)
EPC Dwelling Carbon Dioxide Emission Rate (DER)			-1.1800 (273)

13a. Primary energy - Individual heating systems including micro-CHP

	Energy kWh/year	Primary energy factor kg CO2/kWh	Primary energy kWh/year
Space heating - main system 1	304.2103	1.5837	481.7643 (275)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	1198.8679	1.5207	1823.0881 (278)
Space and water heating			2304.8524 (279)
Pumps, fans and electric keep-hot	136.0977	1.5128	205.8887 (281)
Energy for lighting	185.4911	1.5338	284.5125 (282)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1043.5990	1.4927	-1557.7775
PV Unit electricity exported	-1694.1751	0.4461	-755.7073
Total			-2313.4848 (283)
Total Primary energy kWh/year			481.7688 (286)
Dwelling Primary energy Rate (DPER)			6.8700 (287)

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022) CALCULATION OF TARGET EMISSIONS

1. Overall dwelling characteristics

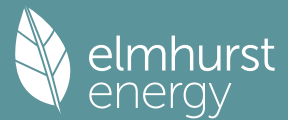
	Area (m2)	Storey height (m)	Volume (m3)
Main dwelling	70.1100 (1b)	x 2.6300 (2b)	= 184.3893 (1b) - (3b)
Ground floor			(4)
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100		
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) =	184.3893 (5)

2. Ventilation rate

	m3 per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 = 0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	3 * 10 = 30.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	30.0000 / (5) = 0.1627 (8)
Pressure test	Yes
Pressure Test Method	Blower Door
Measured/design AP50	5.0000 (17)
Infiltration rate	0.4127 (18)
Number of sides sheltered	0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) = 0.4127 (21)

Wind speed	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)

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Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.5262	0.5159	0.5056	0.4540	0.4437	0.3921	0.3921	0.3817	0.4127	0.4437	0.4643	0.4849 (22b)
Effective ac	0.6384	0.6331	0.6278	0.6030	0.5984	0.5769	0.5769	0.5729	0.5852	0.5984	0.6078	0.6176 (25)

3. Heat losses and heat loss parameter

Element	Gross m2	Openings m2	NetArea m2	U-value W/m2K	A x U W/K	K-value kJ/m2K	A x K kJ/K
Main dwelling							
TER Opaque door			2.1200	1.0000	2.1200		(26)
TER Opening Type			15.4100	1.1450	17.6450		(27)
Heatloss Floor 1			70.1100	0.1300	9.1143		(28a)
External Wall 1	81.6400	17.5300	64.1100	0.1800	11.5398		(29a)
External Roof 1	72.5800		72.5800	0.1100	7.9838		(30)
Total net area of external elements Aum(A, m2)			224.3300				(31)
Fabric heat loss, W/K = Sum (A x U)				(26)...(30) + (32) =	48.4029		(33)

Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K

143.2041 (35)

List of Thermal Bridges

K1 Element	Length	Psi-value	Total
E2 Other lintels (including other steel lintels)	13.0700	0.0500	0.6535
E4 Jamb	32.8000	0.0500	1.6400
E5 Ground floor (normal)	35.9600	0.1600	5.7536
E3 Sill	4.9700	0.0500	0.2485
E11 Eaves (insulation at rafter level)	24.2600	0.0400	0.9704
E13 Gable (insulation at rafter level)	11.8000	0.0800	0.9440
E16 Corner (normal)	8.9200	0.0900	0.8028

Thermal bridges (Sum(L x Psi) calculated using Appendix K)

11.0128 (36)

Point Thermal bridges

(36a) = 0.0000

Total fabric heat loss

(33) + (36) + (36a) = 59.4157 (37)

Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)

(38)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Heat transfer coeff	38.8480	38.5209	38.2003	36.6943	36.4125	35.1009	35.1009	34.8580	35.6061	36.4125	36.9825	37.5785 (38)
Average = Sum(39)m / 12 =	98.2638	97.9367	97.6160	96.1100	95.8283	94.5166	94.5166	94.2737	95.0218	95.8283	96.3983	96.9942 (39)

HLP	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
HLP (average)	1.4016	1.3969	1.3923	1.3708	1.3668	1.3481	1.3481	1.3447	1.3553	1.3668	1.3750	1.3835 (40)
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31

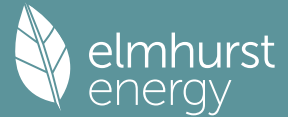
4. Water heating energy requirements (kWh/year)

Assumed occupancy												2.2488 (42)
Hot water usage for mixer showers												
79.5240	78.3289	76.5874	73.2554	70.7965	68.0543	66.4956	68.2239	70.1185	73.0627	76.4663	79.2193 (42a)	
Hot water usage for baths												
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (42b)	
Hot water usage for other uses												
37.6599	36.2904	34.9210	33.5515	32.1821	30.8126	30.8126	32.1821	33.5515	34.9210	36.2904	37.6599 (42c)	
Average daily hot water use (litres/day)												107.5547 (43)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Daily hot water use												
117.1839	114.6193	111.5084	106.8070	102.9786	98.8669	97.3083	100.4060	103.6700	107.9837	112.7568	116.8792 (44)	
Energy content	185.5906	163.2164	171.3843	146.2134	138.6289	121.5684	117.6859	124.3236	127.8355	146.5329	160.6427	183.0046 (45)
Energy content (annual)										Total = Sum(45)m =		1786.6272
Distribution loss (46)m = 0.15 x (45)m												
27.8386	24.4825	25.7076	21.9320	20.7943	18.2353	17.6529	18.6485	19.1753	21.9799	24.0964	27.4507 (46)	
Water storage loss:												
Store volume												150.0000 (47)
a) If manufacturer declared loss factor is known (kWh/day):												1.3938 (48)
Temperature factor from Table 2b												0.5400 (49)
Enter (49) or (54) in (55)												0.7527 (55)
Total storage loss												
23.3325	21.0745	23.3325	22.5798	23.3325	22.5798	23.3325	23.3325	22.5798	23.3325	22.5798	23.3325 (56)	
If cylinder contains dedicated solar storage												
23.3325	21.0745	23.3325	22.5798	23.3325	22.5798	23.3325	23.3325	22.5798	23.3325	22.5798	23.3325 (57)	
Primary loss	23.2624	21.0112	23.2624	22.5120	23.2624	22.5120	23.2624	23.2624	22.5120	23.2624	22.5120	23.2624 (59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)	
Total heat required for water heating calculated for each month												
232.1855	205.3021	217.9792	191.3052	185.2238	166.6603	164.2808	170.9185	172.9273	193.1278	205.7345	229.5995 (62)	
WWHRS	-36.3551	-32.1527	-33.6685	-27.8788	-25.9821	-22.2330	-20.8399	-22.1612	-23.0031	-27.1182	-30.7216	-35.6818 (63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)
Output from w/h												
195.8305	173.1494	184.3107	163.4264	159.2418	144.4272	143.4409	148.7574	149.9242	166.0096	175.0129	193.9177 (64)	
12Total per year (kWh/year)												1997.4487 (64)
Electric shower(s)												1997 (64)
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)	
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =												0.0000 (64a)
Heat gains from water heating, kWh/month												
98.9848	87.9380	94.2612	84.6894	83.3700	76.4950	76.4065	78.6135	78.5788	85.9981	89.4872	98.1249 (65)	

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(66)m	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396 (66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	99.6804	110.3605	99.6804	103.0031	99.6804	103.0031	99.6804	99.6804	103.0031	99.6804	103.0031	99.6804 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	197.5537	199.6037	194.4377	183.4401	169.5576	156.5101	147.7935	145.7435	150.9094	161.9070	175.7895	188.8371 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440 (69)
Pumps, fans	3.0000	3.0000	3.0000	3.0000	3.0000	0.0000	0.0000	0.0000	0.0000	3.0000	3.0000	3.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)												

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	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517 (71)
Water heating gains (Table 5)	133.0441	130.8602	126.6952	117.6242	112.0565	106.2430	102.6969	105.6633	109.1372	115.5889	124.2877	131.8884 (72)
Total internal gains	490.0101	500.5562	480.5452	463.7993	441.0265	422.4881	406.9027	407.8191	419.7815	436.9081	462.8122	480.1377 (73)

6. Solar gains

[Jan]	Area		Solar flux		g		FF		Access		Gains	
	m2		Table 6a		Specific data		Specific data		factor		W	
			W/m2		or Table 6b		or Table 6c		Table 6d			
North	0.6400		10.6334		0.6300		0.7000		0.7700		2.0798 (74)	
East	1.3600		19.6403		0.6300		0.7000		0.7700		8.1632 (76)	
South	2.5200		46.7521		0.6300		0.7000		0.7700		36.0059 (78)	
West	10.8900		19.6403		0.6300		0.7000		0.7700		65.3653 (80)	
Solar gains	111.6141	206.7801	318.7484	441.2193	526.4730	534.1992	510.4216	446.8157	362.0875	239.0091	136.9262	93.3124 (83)
Total gains	601.6241	707.3363	799.2936	905.0186	967.4995	956.6873	917.3242	854.6348	781.8690	675.9172	599.7384	573.4501 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	28.3818	28.4766	28.5701	29.0178	29.1031	29.5070	29.5070	29.5830	29.3501	29.1031	28.9310	28.7533
alpha	2.8921	2.8984	2.9047	2.9345	2.9402	2.9671	2.9671	2.9722	2.9567	2.9402	2.9287	2.9169
util living area	0.9645	0.9435	0.9073	0.8306	0.7148	0.5611	0.4286	0.4717	0.6782	0.8689	0.9452	0.9687 (86)
MIT	18.7096	19.0340	19.5060	20.1049	20.5651	20.8508	20.9502	20.9332	20.7261	20.1082	19.3193	18.6701 (87)
Th 2	19.7621	19.7657	19.7692	19.7858	19.7889	19.8033	19.8033	19.8060	19.7978	19.7889	19.7826	19.7760 (88)
util rest of house	0.9574	0.9325	0.8889	0.7971	0.6587	0.4767	0.3217	0.3627	0.5987	0.8350	0.9326	0.9624 (89)
MIT 2	17.1483	17.5567	18.1451	18.8786	19.4034	19.7029	19.7818	19.7747	19.5922	18.9050	17.9323	17.1068 (90)
Living area fraction	17.6928	18.0719	18.6197	19.3063	19.8085	20.1033	20.1893	20.1787	19.9876	19.3246	18.4160	17.6520 (92)
MIT	17.6928	18.0719	18.6197	19.3063	19.8085	20.1033	20.1893	20.1787	19.9876	19.3246	18.4160	17.6520 (92)
Temperature adjustment	17.6928	18.0719	18.6197	19.3063	19.8085	20.1033	20.1893	20.1787	19.9876	19.3246	18.4160	17.6520 (93)
adjusted MIT	17.6928	18.0719	18.6197	19.3063	19.8085	20.1033	20.1893	20.1787	19.9876	19.3246	18.4160	17.6520 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Utilisation	0.9408	0.9124	0.8675	0.7818	0.6597	0.4990	0.3574	0.3982	0.6125	0.8193	0.9137	0.9469 (94)
Useful gains	565.9788	645.4025	693.3853	707.5662	638.2250	477.4217	327.8549	340.3226	478.8675	553.7921	547.9768	543.0173 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	1316.0286	1290.0154	1183.0783	1000.1487	777.0238	520.1490	339.2479	356.2296	559.4550	836.0643	1090.8405	1304.7616 (97)
Space heating kWh	558.0371	433.1799	364.3316	210.6594	103.2663	0.0000	0.0000	0.0000	0.0000	210.0106	390.8619	566.7377 (98a)
Space heating requirement - total per year (kWh/year)												2837.0844
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	558.0371	433.1799	364.3316	210.6594	103.2663	0.0000	0.0000	0.0000	0.0000	210.0106	390.8619	566.7377 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												2837.0844
Space heating per m2										(98c) / (4) =		40.4662 (99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)
Fraction of space heat from main system(s)												1.0000 (202)
Efficiency of main space heating system 1 (in %)												92.3000 (206)
Efficiency of main space heating system 2 (in %)												0.0000 (207)
Efficiency of secondary/supplementary heating system, %												0.0000 (208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Space heating requirement	558.0371	433.1799	364.3316	210.6594	103.2663	0.0000	0.0000	0.0000	0.0000	210.0106	390.8619	566.7377 (98)
Space heating efficiency (main heating system 1)	92.3000	92.3000	92.3000	92.3000	92.3000	0.0000	0.0000	0.0000	0.0000	92.3000	92.3000	92.3000 (210)
Space heating fuel (main heating system)	604.5905	469.3173	394.7255	228.2334	111.8811	0.0000	0.0000	0.0000	0.0000	227.5304	423.4690	614.0171 (211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating												
Water heating requirement	195.8305	173.1494	184.3107	163.4264	159.2418	144.4272	143.4409	148.7574	149.9242	166.0096	175.0129	193.9177 (64)
Efficiency of water heater (217)m	86.2725	86.0292	85.5584	84.6302	83.1137	79.8000	79.8000	79.8000	79.8000	84.5882	85.8070	79.8000 (216)
Fuel for water heating, kWh/month	226.9907	201.2681	215.4210	193.1065	191.5950	180.9865	179.7505	186.4127	187.8749	196.2563	203.9612	224.6545 (219)
Space cooling fuel requirement (221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (221)
Pumps and Fa	7.3041	6.5973	7.3041	7.0685	7.3041	7.0685	7.3041	7.3041	7.0685	7.3041	7.0685	7.3041 (231)
Lighting	20.7116	16.6156	14.9605	10.9607	8.4664	6.9171	7.7233	10.0391	13.0397	17.1089	19.3244	21.2873 (232)
Electricity generated by PVs (Appendix M) (negative quantity) (233a)m	-54.4906	-71.7980	-96.5686	-101.4298	-103.7749	-94.9831	-93.8069	-91.1847	-85.9084	-78.5268	-58.0922	-47.7255 (233a)
Electricity generated by wind turbines (Appendix M) (negative quantity) (234a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												

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(235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)													
(235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235c)
Electricity generated by PVs (Appendix M) (negative quantity)													
(233b)m	-48.1643	-98.6425	-191.3304	-280.8120	-365.2151	-364.7515	-360.4279	-307.8661	-229.3471	-138.7214	-63.5035	-38.2911	(233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)													
(234b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)													
(235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)													
(235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235d)
Annual totals kWh/year													
Space heating fuel - main system 1												3073.7643	(211)
Space heating fuel - main system 2												0.0000	(213)
Space heating fuel - secondary												0.0000	(215)
Efficiency of water heater												79.8000	
Water heating fuel used												2388.2780	(219)
Space cooling fuel												0.0000	(221)
Electricity for pumps and fans:													
Total electricity for the above, kWh/year												86.0000	(231)
Electricity for lighting (calculated in Appendix L)												167.1546	(232)
Energy saving/generation technologies (Appendices M ,N and Q)													
PV generation												-3465.3624	(233)
Wind generation												0.0000	(234)
Hydro-electric generation (Appendix N)												0.0000	(235a)
Electricity generated - Micro CHP (Appendix N)												0.0000	(235)
Appendix Q - special features													
Energy saved or generated												-0.0000	(236)
Energy used												0.0000	(237)
Total delivered energy for all uses												2249.8345	(238)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year	
Space heating - main system 1	3073.7643	0.2100	645.4905	(261)
Total CO2 associated with community systems			0.0000	(373)
Water heating (other fuel)	2388.2780	0.2100	501.5384	(264)
Space and water heating			1147.0289	(265)
Pumps, fans and electric keep-hot	86.0000	0.1387	11.9293	(267)
Energy for lighting	167.1546	0.1443	24.1256	(268)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-978.2895	0.1360	-133.0823	
PV Unit electricity exported	-2487.0729	0.1266	-314.8438	
Total			-447.9261	(269)
Total CO2, kg/year			735.1576	(272)
EPC Target Carbon Dioxide Emission Rate (TER)			10.4900	(273)

13a. Primary energy - Individual heating systems including micro-CHP

	Energy kWh/year	Primary energy factor kg CO2/kWh	Primary energy kWh/year	
Space heating - main system 1	3073.7643	1.1300	3473.3536	(275)
Total CO2 associated with community systems			0.0000	(473)
Water heating (other fuel)	2388.2780	1.1300	2698.7542	(278)
Space and water heating			6172.1078	(279)
Pumps, fans and electric keep-hot	86.0000	1.5128	130.1008	(281)
Energy for lighting	167.1546	1.5338	256.3874	(282)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-978.2895	1.5028	-1470.2158	
PV Unit electricity exported	-2487.0729	0.4647	-1155.7705	
Total			-2625.9864	(283)
Total Primary energy kWh/year			3932.6096	(286)
Target Primary Energy Rate (TPER)			56.0900	(287)

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022) CALCULATION OF FABRIC ENERGY EFFICIENCY

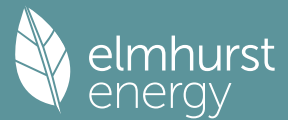
1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)	
Main dwelling	70.1100	2.6300	184.3893	(1b) - (3b)
Ground floor				(4)
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100			
Dwelling volume				(3a)+(3b)+(3c)+(3d)+(3e)...(3n) = 184.3893 (5)

2. Ventilation rate

	m3 per hour	
Number of open chimneys	0 * 80 =	0.0000 (6a)
Number of open flues	0 * 20 =	0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 =	0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 =	0.0000 (6d)
Number of flues attached to other heater	0 * 35 =	0.0000 (6e)
Number of blocked chimneys	0 * 20 =	0.0000 (6f)

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Number of intermittent extract fans 3 * 10 = 30.0000 (7a)
Number of passive vents 0 * 10 = 0.0000 (7b)
Number of flueless gas fires 0 * 40 = 0.0000 (7c)

Air changes per hour
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) = 30.0000 / (5) = 0.1627 (8)
Pressure test Yes
Pressure Test Method Blower Door
Measured/design AP50 1.0000 (17)
Infiltration rate 0.2127 (18)
Number of sides sheltered 0 (19)
Shelter factor (20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor (21) = (18) x (20) = 0.2127 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.2712	0.2659	0.2606	0.2340	0.2287	0.2021	0.2021	0.1967	0.2127	0.2287	0.2393	0.2499 (22b)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.0000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												0.0000 (23c)
Effective ac	0.5368	0.5353	0.5339	0.5274	0.5261	0.5204	0.5204	0.5194	0.5226	0.5261	0.5286	0.5312 (25)

3. Heat losses and heat loss parameter

Element	Gross m2	Openings m2	NetArea m2	U-value W/m2K	A x U W/K	K-value kJ/m2K	A x K kJ/K
Main dwelling							
Door			2.1200	1.0000	2.1200		(26)
Window			20.5800	1.1450	23.5649		(27)
Heatloss Floor 1			70.1100	0.1200	8.4132	110.0000	7712.1000 (28a)
External Wall 1	81.6400	22.7000	58.9400	0.1600	9.4304	9.0000	530.4600 (29a)
External Roof 1	72.5800		72.5800	0.1400	10.1612	9.0000	653.2200 (30)
Total net area of external elements Aum(A, m2)			224.3300				(31)
Fabric heat loss, W/K = Sum (A x U)					(26)...(30) + (32) = 53.6897		(33)
Main dwelling							
Internal Wall 1			127.1400			9.0000	1144.2600 (32c)

Heat capacity Cm = Sum(A x k)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K (28)...(30) + (32) + (32a)...(32e) = 10040.0400 (34)
List of Thermal Bridges 143.2041 (35)

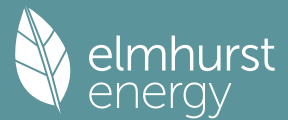
	Length	Psi-value	Total
K1 Element	13.0700	0.0840	1.0979
E2 Other lintels (including other steel lintels)	32.8000	0.0430	1.4104
E4 Jamb	35.9600	0.0210	0.7552
E5 Ground floor (normal)	4.9700	0.0340	0.1690
E3 Sill	24.2600	0.0400	0.9704
E11 Eaves (insulation at rafter level)	11.8000	0.0240	0.2832
E13 Gable (insulation at rafter level)	8.9200	0.0300	0.2676
E16 Corner (normal)			
Thermal bridges (Sum(L x Psi) calculated using Appendix K)			4.9536 (36)
Point Thermal bridges			(36a) = 0.0000
Total fabric heat loss			(33) + (36) + (36a) = 58.6433 (37)

Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)
(38)m Jan 32.6618 Feb 32.5749 Mar 32.4897 Apr 32.0897 May 32.0149 Jun 31.6665 Jul 31.6665 Aug 31.6019 Sep 31.8007 Oct 32.0149 Nov 32.1663 Dec 32.3246 (38)
Heat transfer coeff Jan 91.3051 Feb 91.2182 Mar 91.1330 Apr 90.7330 May 90.6582 Jun 90.3098 Jul 90.3098 Aug 90.2452 Sep 90.4440 Oct 90.6582 Nov 90.8096 Dec 90.9679 (39)
Average = Sum(39)m / 12 = 90.7327
HLP Jan 1.3023 Feb 1.3011 Mar 1.2999 Apr 1.2942 May 1.2931 Jun 1.2881 Jul 1.2881 Aug 1.2872 Sep 1.2900 Oct 1.2931 Nov 1.2952 Dec 1.2975 (40)
HLP (average) 1.2941
Days in mont 31 28 31 30 31 30 31 31 30 31 30 31

4. Water heating energy requirements (kWh/year)

Assumed occupancy 2.2488 (42)
Hot water usage for mixer showers 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (42a)
Hot water usage for baths 26.7528 26.3555 25.7960 24.7643 23.9919 23.1353 22.6727 23.2283 23.8332 24.7497 25.8026 26.6623 (42b)
Hot water usage for other uses 37.6599 36.2904 34.9210 33.5515 32.1821 30.8126 30.8126 32.1821 33.5515 34.9210 36.2904 37.6599 (42c)
Average daily hot water use (litres/day) 59.0403 (43)
Daily hot water use Jan 64.4127 Feb 62.6459 Mar 60.7170 Apr 58.3159 May 56.1740 Jun 53.9480 Jul 53.4853 Aug 55.4103 Sep 57.3847 Oct 59.6707 Nov 62.0930 Dec 64.3222 (44)
Energy conte 102.0139 89.2069 93.3197 79.8315 75.6209 66.3353 64.6858 68.6096 70.7611 80.9726 88.4629 100.7131 (45)
Energy content (annual) Total = Sum(45)m = 980.5334
Distribution loss (46)m = 0.15 x (45)m 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (46)
Water storage loss:
Total storage loss 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (56)
If cylinder contains dedicated solar storage 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (57)
Primary loss 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (59)
Combi loss 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (61)
Total heat required for water heating calculated for each month 86.7118 75.8259 79.3217 67.8568 64.2778 56.3850 54.9830 58.3181 60.1469 68.8267 75.1935 85.6061 (62)
WWHRS 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (63a)
PV diverter 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (63b)
Solar input 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (63c)
FGHRS 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 (63d)
Output from w/h 86.7118 75.8259 79.3217 67.8568 64.2778 56.3850 54.9830 58.3181 60.1469 68.8267 75.1935 85.6061 (64)
Total per year (kWh/year) = Sum(64)m = 833.4534 (64)
Electric shower(s) 833 (64)
49.5912 44.1862 48.2496 46.0440 46.9080 44.7457 46.2372 46.9080 46.0440 48.2496 47.3423 49.5912 (64a)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m = 564.0971 (64a)

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Heat gains from water heating, kWh/month
34.0758 30.0030 31.8928 28.4752 27.7965 25.2827 25.3050 26.3065 26.5477 29.2691 30.6339 33.7993 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(66)m	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	(66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	99.6431	110.3192	99.6431	102.9646	99.6431	102.9646	99.6431	99.6431	102.9646	99.6431	102.9646	99.6431	(67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	197.5537	199.6037	194.4377	183.4401	169.5576	156.5101	147.7935	145.7435	150.9094	161.9070	175.7895	188.8371	(68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	(69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(70)
Losses e.g. evaporation (negative values) (Table 5)	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	(71)
Water heating gains (Table 5)	45.8008	44.6473	42.8667	39.5489	37.3608	35.1148	34.0122	35.3582	36.8719	39.3402	42.5472	45.4292	(72)
Total internal gains	399.7294	411.3021	393.6795	382.6855	363.2935	351.3213	338.1806	337.4767	347.4777	357.6222	378.0331	390.6413	(73)

6. Solar gains

[Jan]				Area	Solar flux		g		FF		Access		Gains		
				m2	Table 6a		Specific data		Specific data		factor		W		
					W/m2		or Table 6b		or Table 6c		Table 6d				
North				0.8500	10.6334		0.7600		0.7000		0.7700		3.3322 (74)		
East				1.8100	19.6403		0.7600		0.7000		0.7700		13.1060 (76)		
South				3.3600	46.7521		0.7600		0.7000		0.7700		57.9142 (78)		
West				14.5600	19.6403		0.7600		0.7000		0.7700		105.4274 (80)		
Solar gains				179.7799	333.0931	513.5087	710.8654	848.2508	860.7078	822.3937	719.8947	583.3490	385.0244	220.5561	150.2972 (83)
Total gains				579.5093	744.3952	907.1882	1093.5509	1211.5443	1212.0292	1160.5743	1057.3714	930.8267	742.6466	598.5891	540.9385 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)													21.0000 (85)	
Utilisation factor for gains for living area, nil,m (see Table 9a)														
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
tau	30.5448	30.5739	30.6025	30.7374	30.7628	30.8815	30.8815	30.9036	30.8357	30.7628	30.7115	30.6581		
alpha	3.0363	3.0383	3.0402	3.0492	3.0509	3.0588	3.0588	3.0602	3.0557	3.0509	3.0474	3.0439		
util living area	0.9665	0.9331	0.8721	0.7545	0.6049	0.4496	0.3338	0.3786	0.5893	0.8346	0.9431	0.9722	(86)	
MIT	18.8624	19.2844	19.8243	20.3943	20.7552	20.9265	20.9778	20.9672	20.8321	20.2825	19.4511	18.7762	(87)	
Th 2	19.8390	19.8400	19.8409	19.8454	19.8463	19.8502	19.8502	19.8509	19.8487	19.8463	19.8446	19.8428	(88)	
util rest of house	0.9599	0.9209	0.8498	0.7157	0.5488	0.3774	0.2501	0.2896	0.5126	0.7967	0.9304	0.9667	(89)	
MIT 2	17.9278	18.3387	18.8545	19.3780	19.6828	19.8130	19.8428	19.8392	19.7536	19.2985	18.5117	17.8459	(90)	
Living area fraction										fLA = Living area / (4) =				0.3487 (91)
MIT	18.2538	18.6685	19.1927	19.7324	20.0568	20.2013	20.2387	20.2326	20.1297	19.6417	18.8393	18.1703	(92)	
Temperature adjustment													0.0000	
adjusted MIT	18.2538	18.6685	19.1927	19.7324	20.0568	20.2013	20.2387	20.2326	20.1297	19.6417	18.8393	18.1703	(93)	

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Utilisation	0.9480	0.9054	0.8352	0.7115	0.5596	0.4001	0.2788	0.3198	0.5324	0.7890	0.9164	0.9559	(94)
Useful gains	549.3640	674.0031	757.6864	778.0734	677.9443	484.8943	323.6112	338.1730	495.5736	585.9274	548.5346	517.1063	(95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000	(96)
Heat loss rate W	1274.0487	1255.9383	1156.7216	982.8569	757.6105	505.8526	328.6059	345.8707	545.3515	819.7007	1066.0439	1270.8490	(97)
Space heating kWh	539.1654	391.0604	296.8822	147.4441	59.2716	0.0000	0.0000	0.0000	0.0000	173.9273	372.6067	560.7846	(98a)
Space heating requirement - total per year (kWh/year)												2541.1423	
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(98b)
Solar heating contribution - total per year (kWh/year)												0.0000	
Space heating kWh	539.1654	391.0604	296.8822	147.4441	59.2716	0.0000	0.0000	0.0000	0.0000	173.9273	372.6067	560.7846	(98c)
Space heating requirement after solar contribution - total per year (kWh/year)												2541.1423	
Space heating per m2										(98c) / (4) =		36.2451	(99)

8c. Space cooling requirement

Calculated for June, July and August. See Table 10b	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Ext. temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000	
Heat loss rate W	0.0000	0.0000	0.0000	0.0000	0.0000	848.9118	668.2922	685.8638	0.0000	0.0000	0.0000	0.0000	(100)
Utilisation	0.0000	0.0000	0.0000	0.0000	0.0000	0.8987	0.9344	0.9141	0.0000	0.0000	0.0000	0.0000	(101)
Useful loss	0.0000	0.0000	0.0000	0.0000	0.0000	762.9444	624.4565	626.9587	0.0000	0.0000	0.0000	0.0000	(102)
Total gains	0.0000	0.0000	0.0000	0.0000	0.0000	1379.1698	1320.9767	1202.2837	0.0000	0.0000	0.0000	0.0000	(103)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	443.6823	518.2110	428.0418	0.0000	0.0000	0.0000	0.0000	(104)
Cooled fraction									fc = cooled area / (4) =			1.0000	(105)
Intermittency factor (Table 10b)	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	(106)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	110.9206	129.5528	107.0104	0.0000	0.0000	0.0000	0.0000	(107)
Space cooling requirement												347.4838	(107)
Energy for space heating												36.2451	(99)

Full SAP Calculation Printout



Energy for space cooling 4.9563 (108)
 Total 41.2013 (109)
 Fabric Energy Efficiency (DFEE) 41.2 (109)

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
 CALCULATION OF TARGET FABRIC ENERGY EFFICIENCY

1. Overall dwelling characteristics

	Area (m ²)	Storey height (m)	Volume (m ³)
Main dwelling	70.1100 (1b)	x 2.6300 (2b)	= 184.3893 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100		(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n)	= 184.3893 (5)

2. Ventilation rate

	m ³ per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 = 0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	3 * 10 = 30.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	30.0000 / (5) = 0.1627 (8)
Pressure test	Yes
Pressure Test Method	Blower Door
Measured/design AP50	5.0000 (17)
Infiltration rate	0.4127 (18)
Number of sides sheltered	0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) = 0.4127 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.5262	0.5159	0.5056	0.4540	0.4437	0.3921	0.3921	0.3817	0.4127	0.4437	0.4643	0.4849 (22b)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.0000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												0.0000 (23c)
Effective ac	0.6384	0.6331	0.6278	0.6030	0.5984	0.5769	0.5769	0.5729	0.5852	0.5984	0.6078	0.6176 (25)

3. Heat losses and heat loss parameter

Element	Gross m ²	Openings m ²	NetArea m ²	U-value W/m ² K	A x U W/K	K-value kJ/m ² K	A x K kJ/K
Main dwelling							
TER Opaque door			2.1200	1.0000	2.1200		(26)
TER Opening Type			15.4100	1.1450	17.6450		(27)
Heatloss Floor 1			70.1100	0.1300	9.1143		(28a)
External Wall 1	81.6400	17.5300	64.1100	0.1800	11.5398		(29a)
External Roof 1	72.5800		72.5800	0.1100	7.9838		(30)
Total net area of external elements Aum(A, m ²)			224.3300				(31)
Fabric heat loss, W/K = Sum (A x U)			(26)...(30) + (32) =		48.4029		(33)

Thermal mass parameter (TMP = Cm / TFA) in kJ/m²K 143.2041 (35)

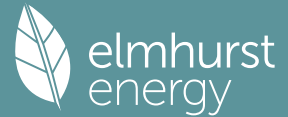
List of Thermal Bridges	Length	Psi-value	Total
K1 Element			
E2 Other lintels (including other steel lintels)	13.0700	0.0500	0.6535
E4 Jamb	32.8000	0.0500	1.6400
E5 Ground floor (normal)	35.9600	0.1600	5.7536
E3 Sill	4.9700	0.0500	0.2485
E11 Eaves (insulation at rafter level)	24.2600	0.0400	0.9704
E13 Gable (insulation at rafter level)	11.8000	0.0800	0.9440
E16 Corner (normal)	8.9200	0.0900	0.8028
Thermal bridges (Sum(L x Psi) calculated using Appendix K)			11.0128 (36)
Point Thermal bridges			(36a) = 0.0000
Total fabric heat loss			(33) + (36) + (36a) = 59.4157 (37)

Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(38)m	38.8480	38.5209	38.2003	36.6943	36.4125	35.1009	35.1009	34.8580	35.6061	36.4125	36.9825	37.5785 (38)
Heat transfer coeff	98.2638	97.9367	97.6160	96.1100	95.8283	94.5166	94.5166	94.2737	95.0218	95.8283	96.3983	96.9942 (39)
Average = Sum(39)m / 12 =												96.1087

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
HLP	1.4016	1.3969	1.3923	1.3708	1.3668	1.3481	1.3481	1.3447	1.3553	1.3668	1.3750	1.3835 (40)
HLP (average)												1.3708
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31

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4. Water heating energy requirements (kWh/year)

Assumed occupancy												2.2488 (42)
Hot water usage for mixer showers												0.0000 (42a)
Hot water usage for baths	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (42b)
Hot water usage for other uses	26.7528	26.3555	25.7960	24.7643	23.9919	23.1353	22.6727	23.2283	23.8332	24.7497	25.8026	26.6623 (42b)
Average daily hot water use (litres/day)	37.6599	36.2904	34.9210	33.5515	32.1821	30.8126	30.8126	32.1821	33.5515	34.9210	36.2904	37.6599 (42c)
												59.0403 (43)
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Energy content (annual)	64.4127	62.6459	60.7170	58.3159	56.1740	53.9480	53.4853	55.4103	57.3847	59.6707	62.0930	64.3222 (44)
Distribution loss (46)m = 0.15 x (45)m	102.0139	89.2069	93.3197	79.8315	75.6209	66.3353	64.6858	68.6096	70.7611	80.9726	88.4629	100.7131 (45)
												980.5334
Water storage loss:												0.0000 (46)
Total storage loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (56)
If cylinder contains dedicated solar storage	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (57)
Primary loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
Total heat required for water heating calculated for each month	86.7118	75.8259	79.3217	67.8568	64.2778	56.3850	54.9830	58.3181	60.1469	68.8267	75.1935	85.6061 (62)
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
PV diverter	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63b)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)
Output from w/h	86.7118	75.8259	79.3217	67.8568	64.2778	56.3850	54.9830	58.3181	60.1469	68.8267	75.1935	85.6061 (64)
												833.4534 (64)
												833 (64)
12Total per year (kWh/year)												85.6061 (64a)
Electric shower(s)	49.5912	44.1862	48.2496	46.0440	46.9080	44.7457	46.2372	46.9080	46.0440	48.2496	47.3423	49.5912 (64a)
												564.0971 (64a)
Heat gains from water heating, kWh/month	34.0758	30.0030	31.8928	28.4752	27.7965	25.2827	25.3050	26.3065	26.5477	29.2691	30.6339	33.7993 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(66)m	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396	112.4396 (66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	99.6804	110.3605	99.6804	103.0031	99.6804	103.0031	99.6804	99.6804	103.0031	99.6804	103.0031	99.6804 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	197.5537	199.6037	194.4377	183.4401	169.5576	156.5101	147.7935	145.7435	150.9094	161.9070	175.7895	188.8371 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440	34.2440 (69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517 (71)
Water heating gains (Table 5)	45.8008	44.6473	42.8667	39.5489	37.3608	35.1148	34.0122	35.3582	36.8719	39.3402	42.5472	45.4292 (72)
Total internal gains	399.7667	411.3433	393.7168	382.7240	363.3308	351.3599	338.2179	337.5140	347.5162	357.6594	378.0716	390.6786 (73)

6. Solar gains

[Jan]					Area m2	Solar flux Table 6a W/m2	g Specific data or Table 6b	FF Specific data or Table 6c	Access factor Table 6d			Gains W
North					0.6400	10.6334	0.6300	0.7000	0.7700			2.0798 (74)
East					1.3600	19.6403	0.6300	0.7000	0.7700			8.1632 (76)
South					2.5200	46.7521	0.6300	0.7000	0.7700			36.0059 (78)
West					10.8900	19.6403	0.6300	0.7000	0.7700			65.3653 (80)
Solar gains	111.6141	206.7801	318.7484	441.2193	526.4730	534.1992	510.4216	446.8157	362.0875	239.0091	136.9262	93.3124 (83)
Total gains	511.3808	618.1235	712.4652	823.9433	889.8038	885.5591	848.6395	784.3297	709.6037	596.6685	514.9978	483.9910 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	28.3818	28.4766	28.5701	29.0178	29.1031	29.5070	29.5070	29.5830	29.3501	29.1031	28.9310	28.7533
alpha	2.8921	2.8984	2.9047	2.9345	2.9402	2.9671	2.9671	2.9722	2.9567	2.9402	2.9287	2.9169
util living area	0.9761	0.9586	0.9274	0.8571	0.7468	0.5935	0.4581	0.5060	0.7170	0.8976	0.9617	0.9794 (86)
MIT	18.5345	18.8723	19.3666	20.0060	20.5061	20.8262	20.9406	20.9193	20.6791	19.9956	19.1636	18.4934 (87)
Th 2	19.7621	19.7657	19.7692	19.7858	19.7889	19.8033	19.8033	19.8060	19.7978	19.7889	19.7826	19.7760 (88)
util rest of house	0.9711	0.9502	0.9122	0.8270	0.6927	0.5076	0.3458	0.3919	0.6393	0.8691	0.9524	0.9751 (89)
MIT 2	17.5538	17.8880	18.3724	18.9917	19.4429	19.7108	19.7830	19.7759	19.6060	19.0002	18.1909	17.5225 (90)
Living area fraction	17.8958	18.2313	18.7191	19.3454	19.8137	20.0998	20.1867	20.1747	19.9802	19.3473	18.5301	17.8611 (91)
MIT	17.8958	18.2313	18.7191	19.3454	19.8137	20.0998	20.1867	20.1747	19.9802	19.3473	18.5301	17.8611 (92)
Temperature adjustment	17.8958	18.2313	18.7191	19.3454	19.8137	20.0998	20.1867	20.1747	19.9802	19.3473	18.5301	17.8611 (93)

8. Space heating requirement

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
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Utilisation	0.9607	0.9363	0.8959	0.8142	0.6939	0.5302	0.3833	0.4290	0.6524	0.8563	0.9395	0.9657 (94)
Useful gains	491.2675	578.7243	638.3331	670.8540	617.4542	469.5571	325.2643	336.4531	462.9395	510.9112	483.8623	467.3916 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	1335.9746	1305.6209	1192.7815	1003.9120	777.5207	519.8189	338.9988	355.8524	558.7479	838.2395	1101.8450	1325.0463 (97)
Space heating kWh	628.4621	488.4745	412.5096	239.8017	119.0894	0.0000	0.0000	0.0000	0.0000	243.5323	444.9475	638.0951 (98a)
Space heating requirement - total per year (kWh/year)												3214.9123
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	628.4621	488.4745	412.5096	239.8017	119.0894	0.0000	0.0000	0.0000	0.0000	243.5323	444.9475	638.0951 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												3214.9123
Space heating per m2												(98c) / (4) = 45.8553 (99)

8c. Space cooling requirement

Calculated for June, July and August. See Table 10b

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Ext. temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000
Heat loss rate W	0.0000	0.0000	0.0000	0.0000	0.0000	888.4562	699.4230	716.4802	0.0000	0.0000	0.0000	0.0000 (100)
Utilisation	0.0000	0.0000	0.0000	0.0000	0.0000	0.7895	0.8507	0.8204	0.0000	0.0000	0.0000	0.0000 (101)
Useful loss	0.0000	0.0000	0.0000	0.0000	0.0000	701.3989	594.9670	587.7733	0.0000	0.0000	0.0000	0.0000 (102)
Total gains	0.0000	0.0000	0.0000	0.0000	0.0000	997.5397	956.3375	883.1050	0.0000	0.0000	0.0000	0.0000 (103)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	213.2214	268.8597	219.7268	0.0000	0.0000	0.0000	0.0000 (104)
Cooled fraction									fc = cooled area / (4) =			
Intermittency factor (Table 10b)	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500	0.2500 (105)
Space cooling kWh	0.0000	0.0000	0.0000	0.0000	0.0000	53.3053	67.2149	54.9317	0.0000	0.0000	0.0000	0.0000 (107)
Space cooling requirement												175.4520 (107)
Energy for space heating												45.8553 (99)
Energy for space cooling												2.5025 (108)
Total												48.3578 (109)
Fabric Energy Efficiency (TFEE)												48.4 (109)

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF ENERGY RATING

1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)
Main dwelling	70.1100 (1b)	x 2.6300 (2b)	= 184.3893 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100		(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) =	184.3893 (5)

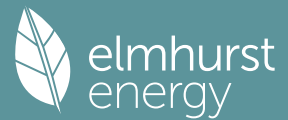
2. Ventilation rate

	m3 per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 = 0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	0 * 10 = 0.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	0.0000 / (5) = 0.0000 (8)
Pressure test	Yes
Pressure Test Method	Blower Door
Measured/design AP50	1.0000 (17)
Infiltration rate	0.0500 (18)
Number of sides sheltered	0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) = 0.0500 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000 (22)
Wind factor	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750 (22a)
Adj infilt rate	0.0638	0.0625	0.0613	0.0550	0.0537	0.0475	0.0475	0.0463	0.0500	0.0537	0.0563	0.0588 (22b)
Balanced mechanical ventilation with heat recovery												0.5000 (23a)
If mechanical ventilation												0.5000 (23b)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												84.6000 (23c)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												
Effective ac	0.1407	0.1395	0.1382	0.1320	0.1307	0.1245	0.1245	0.1232	0.1270	0.1307	0.1332	0.1357 (25)

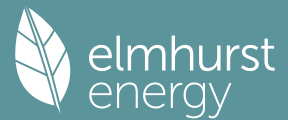
3. Heat losses and heat loss parameter

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Element													
Main dwelling													
Door													(26)
Window													(27)
Heatloss Floor 1													(28a)
External Wall 1													(29a)
External Roof 1													(30)
Total net area of external elements Aum(A, m2)													(31)
Fabric heat loss, W/K = Sum (A x U)													(32)
Main dwelling													(33)
Internal Wall 1													(32c)
Heat capacity Cm = Sum(A x k)													(34)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K													(35)
List of Thermal Bridges													
K1 Element													
E2 Other lintels (including other steel lintels)													
E4 Jamb													
E5 Ground floor (normal)													
E3 Sill													
E11 Eaves (insulation at rafter level)													
E13 Gable (insulation at rafter level)													
E16 Corner (normal)													
Thermal bridges (Sum(L x Psi) calculated using Appendix K)													
Point Thermal bridges													
Total fabric heat loss													
Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)													
(38)m													
Heat transfer coeff													
Average = Sum(39)m / 12 =													
HLP													
HLP (average)													
Days in mont													
4. Water heating energy requirements (kWh/year)													
Assumed occupancy													
Hot water usage for mixer showers													
Hot water usage for baths													
Hot water usage for other uses													
Average daily hot water use (litres/day)													
Daily hot water use													
Energy conte													
Energy content (annual)													
Distribution loss (46)m = 0.15 x (45)m													
Water storage loss:													
Store volume													
a) If manufacturer declared loss factor is known (kWh/day):													
Temperature factor from Table 2b													
Enter (49) or (54) in (55)													
Total storage loss													
If cylinder contains dedicated solar storage													
Primary loss													
Combi loss													
Total heat required for water heating calculated for each month													
WWHRS													
PV diverter													
Solar input													
FGHRS													
Output from w/h													
Electric shower(s)													
Heat gains from water heating, kWh/month													
5. Internal gains (see Table 5 and 5a)													
Metabolic gains (Table 5), Watts													
(66)m													
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5													
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5													
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5													
Pumps, fans													
Losses e.g. evaporation (negative values) (Table 5)													
Water heating gains (Table 5)													
Total internal gains													

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6. Solar gains

[Jan]	Area m2				Solar flux Table 6a W/m2	Specific data or Table 6b	g	Specific data or Table 6c	FF or Table 6c	Access factor Table 6d	Gains W
North	0.8500				10.6334		0.7600		0.7000	0.7700	3.3322 (74)
East	1.8100				19.6403		0.7600		0.7000	0.7700	13.1060 (76)
South	3.3600				46.7521		0.7600		0.7000	0.7700	57.9142 (78)
West	14.5600				19.6403		0.7600		0.7000	0.7700	105.4274 (80)
Solar gains	179.7799	333.0931	513.5087	710.8654	848.2508	860.7078	822.3937	719.8947	583.3490	385.0244	150.2972 (83)
Total gains	740.3870	891.6402	1055.8251	1223.0887	1330.5608	1316.0584	1261.9181	1162.2618	1041.2556	870.4048	697.4686 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	41.4967	41.5437	41.5909	41.8281	41.8759	42.1163	42.1163	42.1648	42.0198	41.8759	41.7804	41.6854
alpha	3.7664	3.7696	3.7727	3.7885	3.7917	3.8078	3.8078	3.8110	3.8013	3.7917	3.7854	3.7790
util living area	0.9176	0.8589	0.7567	0.6059	0.4514	0.3191	0.2302	0.2606	0.4287	0.6930	0.8704	0.9304 (86)
Living	20.1085	20.3423	20.6013	20.7982	20.8844	20.9119	20.9174	20.9166	20.8977	20.7548	20.3973	20.0490
Non living	19.0919	19.3752	19.6797	19.9025	19.9915	20.0211	20.0252	20.0256	20.0084	19.8647	19.4521	19.0219
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.5440	20.3423	20.6013	20.7982	20.8844	20.9119	20.9174	20.9166	20.8977	20.7548	20.3973	20.1820 (87)
Th 2	20.1180	20.1189	20.1198	20.1244	20.1253	20.1298	20.1298	20.1308	20.1280	20.1253	20.1234	20.1216 (88)
util rest of house	0.9054	0.8405	0.7297	0.5711	0.4123	0.2767	0.1850	0.2119	0.3779	0.6525	0.8500	0.9197 (89)
MIT 2	19.7089	19.3752	19.6797	19.9025	19.9915	20.0211	20.0252	20.0256	20.0084	19.8647	19.4521	19.2200 (90)
Living area fraction										fLA = Living area / (4) =		0.3487 (91)
MIT	20.0001	19.7124	20.0011	20.2149	20.3029	20.3318	20.3363	20.3363	20.3186	20.1751	19.7817	19.5555 (92)
Temperature adjustment												0.0000
adjusted MIT	20.0001	19.7124	20.0011	20.2149	20.3029	20.3318	20.3363	20.3363	20.3186	20.1751	19.7817	19.5555 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Utilisation	0.9036	0.8311	0.7259	0.5748	0.4204	0.2866	0.1957	0.2234	0.3892	0.6549	0.8411	0.9118 (94)
Useful gains	669.0450	741.0601	766.4018	702.9900	559.3386	377.1719	247.0132	259.6848	405.2534	570.0503	621.5583	635.9798 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)
Heat loss rate W	1055.1691	994.3835	905.3256	754.4234	572.9464	379.5509	247.4154	260.3603	412.7328	637.6947	846.5225	1027.3375 (97)
Space heating kWh	287.2764	170.2333	103.3594	37.0321	10.1242	0.0000	0.0000	0.0000	0.0000	50.3274	161.9742	291.1701 (98a)
Space heating requirement - total per year (kWh/year)												1111.4972
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	287.2764	170.2333	103.3594	37.0321	10.1242	0.0000	0.0000	0.0000	0.0000	50.3274	161.9742	291.1701 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												1111.4972
Space heating per m2										(98c) / (4) =		15.8536 (99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)
Fraction of space heat from main system(s)												1.0000 (202)
Efficiency of main space heating system 1 (in %)												401.8421 (206)
Efficiency of main space heating system 2 (in %)												0.0000 (207)
Efficiency of secondary/supplementary heating system, %												0.0000 (208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Space heating requirement	287.2764	170.2333	103.3594	37.0321	10.1242	0.0000	0.0000	0.0000	0.0000	50.3274	161.9742	291.1701 (98)
Space heating efficiency (main heating system 1)	401.8421	401.8421	401.8421	401.8421	401.8421	0.0000	0.0000	0.0000	0.0000	401.8421	401.8421	401.8421 (210)
Space heating fuel (main heating system)	71.4899	42.3632	25.7214	9.2156	2.5195	0.0000	0.0000	0.0000	0.0000	12.5242	40.3079	72.4588 (211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating												
Water heating requirement	242.1656	214.3164	227.9593	200.9634	195.2039	176.3184	174.2609	180.8986	182.5855	203.1079	215.3927	239.5796 (64)
Efficiency of water heater	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890 (216)
Fuel for water heating, kWh/month	118.3669	104.7546	111.4230	98.2278	95.4127	86.1818	85.1761	88.4205	89.2450	99.2760	105.2807	117.1028 (219)
Space cooling fuel requirement												
(221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (221)
Pumps and Fa	11.5590	10.4404	11.5590	11.861	11.5590	11.861	11.5590	11.5590	11.861	11.5590	11.861	11.5590 (231)
Lighting	22.9836	18.4383	16.6017	12.1631	9.3951	7.6759	8.5705	11.1403	14.4702	18.9857	21.4443	23.6224 (232)
Electricity generated by PVs (Appendix M) (negative quantity)												
(233a)m	-39.3004	-61.9480	-98.8475	-117.3198	-128.9592	-120.7662	-118.8555	-111.0796	-94.6886	-73.3204	-44.7070	-32.8007 (233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)												
(234a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												
(235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)												
(235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235c)

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Electricity generated by PVs (Appendix M) (negative quantity)													
(233b)m	-16.4103	-42.5615	-106.8536	-193.7202	-282.4430	-293.9538	-286.1186	-225.3275	-143.7754	-68.0991	-23.7218	-12.1965	(233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)													
(234b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)													
(235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)													
(235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(235d)
Annual totals kWh/year													
Space heating fuel - main system 1												276.6005	(211)
Space heating fuel - main system 2												0.0000	(213)
Space heating fuel - secondary												0.0000	(215)
Efficiency of water heater												204.5890	
Water heating fuel used												1198.8679	(219)
Space cooling fuel												0.0000	(221)

Electricity for pumps and fans:													
(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6050)													
mechanical ventilation fans (SFP = 0.6050)												136.0977	(230a)
Total electricity for the above, kWh/year												136.0977	(231)
Electricity for lighting (calculated in Appendix L)												185.4911	(232)

Energy saving/generation technologies (Appendices M ,N and Q)													
PV generation												-2737.7741	(233)
Wind generation												0.0000	(234)
Hydro-electric generation (Appendix N)												0.0000	(235a)
Electricity generated - Micro CHP (Appendix N)												0.0000	(235)
Appendix Q - special features													
Energy saved or generated												-0.0000	(236)
Energy used												0.0000	(237)
Total delivered energy for all uses												-940.7168	(238)

10a. Fuel costs - using Table 12 prices

	Fuel kWh/year	Fuel price p/kWh	Fuel cost £/year	
Space heating - main system 1	276.6005	16.4900	45.6114	(240)
Total CO2 associated with community systems			0.0000	(473)
Water heating (other fuel)	1198.8679	16.4900	197.6933	(247)
Energy for instantaneous electric shower(s)	0.0000	16.4900	0.0000	(247a)
Pumps, fans and electric keep-hot	136.0977	16.4900	22.4425	(249)
Energy for lighting	185.4911	16.4900	30.5875	(250)
Additional standing charges			0.0000	(251)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-1042.5928	16.4900	-171.9236	
PV Unit electricity exported	-1695.1813	5.5900	-94.7606	
Total			-266.6842	(252)
Total energy cost			29.6505	(255)

11a. SAP rating - Individual heating systems

Energy cost deflator (Table 12):		0.3600	(256)
Energy cost factor (ECF)	$[(255) \times (256)] / [(4) + 45.0] =$	0.0927	(257)
SAP value		98.4968	
SAP rating (Section 12)		98	(258)
SAP band		A	

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

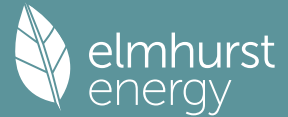
	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year	
Space heating - main system 1	276.6005	0.1577	43.6173	(261)
Total CO2 associated with community systems			0.0000	(373)
Water heating (other fuel)	1198.8679	0.1408	168.8173	(264)
Space and water heating			212.4347	(265)
Pumps, fans and electric keep-hot	136.0977	0.1387	18.8784	(267)
Energy for lighting	185.4911	0.1443	26.7721	(268)
Energy saving/generation technologies				
PV Unit electricity used in dwelling	-1042.5928	0.1333	-138.9873	
PV Unit electricity exported	-1695.1813	0.1216	-206.1390	
Total			-345.1263	(269)
Total CO2, kg/year			-87.0411	(272)
CO2 emissions per m2			-1.2400	(273)
EI value			101.0132	
EI rating			101	(274)
EI band			A	

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF EPC COSTS, EMISSIONS AND PRIMARY ENERGY

1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)	
Main dwelling	70.1100	2.6300	184.3893	(1b) - (3b)
Ground floor				(4)
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100			
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n) =	184.3893	(5)

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2. Ventilation rate

			m3 per hour	
Number of open chimneys	0	* 80 =	0.0000	(6a)
Number of open flues	0	* 20 =	0.0000	(6b)
Number of chimneys / flues attached to closed fire	0	* 10 =	0.0000	(6c)
Number of flues attached to solid fuel boiler	0	* 20 =	0.0000	(6d)
Number of flues attached to other heater	0	* 35 =	0.0000	(6e)
Number of blocked chimneys	0	* 20 =	0.0000	(6f)
Number of intermittent extract fans	0	* 10 =	0.0000	(7a)
Number of passive vents	0	* 10 =	0.0000	(7b)
Number of flueless gas fires	0	* 40 =	0.0000	(7c)
			Air changes per hour	
Infiltration due to chimneys, flues and fans	= (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =		0.0000 / (5) =	0.0000 (8)
Pressure test			Yes	
Pressure Test Method			Blower Door	
Measured/design AP50			1.0000 (17)	
Infiltration rate			0.0500 (18)	
Number of sides sheltered			0 (19)	
Shelter factor	(20) = 1 - [0.075 x (19)] =		1.0000 (20)	
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) =		0.0500 (21)	

Wind speed	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind factor	5.9000	5.5000	5.4000	5.0000	5.0000	4.2000	4.2000	4.1000	4.4000	5.1000	5.3000	5.8000
Adj infilt rate	1.4750	1.3750	1.3500	1.2500	1.2500	1.0500	1.0500	1.0250	1.1000	1.2750	1.3250	1.4500
Balanced mechanical ventilation with heat recovery	0.0738	0.0688	0.0675	0.0625	0.0625	0.0525	0.0525	0.0512	0.0550	0.0638	0.0663	0.0725
If mechanical ventilation												0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												84.6000 (23c)
Effective ac	0.1507	0.1457	0.1445	0.1395	0.1395	0.1295	0.1295	0.1282	0.1320	0.1407	0.1432	0.1495

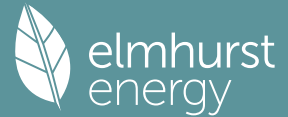
3. Heat losses and heat loss parameter

Element	Gross m2	Openings m2	NetArea m2	U-value W/m2K	A x U W/K	K-value kJ/m2K	A x K kJ/K						
Main dwelling													
Door			2.1200	1.0000	2.1200		(26)						
Window			20.5800	1.1450	23.5649		(27)						
Heatloss Floor 1			70.1100	0.1200	8.4132	110.0000	7712.1000 (28a)						
External Wall 1	81.6400	22.7000	58.9400	0.1600	9.4304	9.0000	530.4600 (29a)						
External Roof 1	72.5800		72.5800	0.1400	10.1612	9.0000	653.2200 (30)						
Total net area of external elements Aum(A, m2)			224.3300				(31)						
Fabric heat loss, W/K = Sum (A x U)			(26)...(30) + (32) =	53.6897			(33)						
Main dwelling													
Internal Wall 1			127.1400			9.0000	1144.2600 (32c)						
Heat capacity Cm = Sum(A x k)				(28)...(30) + (32) + (32a)...(32e) =	10040.0400		(34)						
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K							143.2041 (35)						
List of Thermal Bridges													
K1 Element				Length	Psi-value	Total							
E2 Other lintels (including other steel lintels)				13.0700	0.0840	1.0979							
E4 Jamb				32.8000	0.0430	1.4104							
E5 Ground floor (normal)				35.9600	0.0210	0.7552							
E3 Sill				4.9700	0.0340	0.1690							
E11 Eaves (insulation at rafter level)				24.2600	0.0400	0.9704							
E13 Gable (insulation at rafter level)				11.8000	0.0240	0.2832							
E16 Corner (normal)				8.9200	0.0300	0.2676							
Thermal bridges (Sum(L x Psi) calculated using Appendix K)							4.9536 (36)						
Point Thermal bridges						(36a) =	0.0000						
Total fabric heat loss						(33) + (36) + (36a) =	58.6433 (37)						
Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)													
(38)m	Jan 9.1729	Feb 8.8687	Mar 8.7926	Apr 8.4884	May 8.4884	Jun 7.8799	Jul 7.8799	Aug 7.8038	Sep 8.0320	Oct 8.5644	Nov 8.7165	Dec 9.0968	(38)
Heat transfer coeff	67.8162	67.5120	67.4359	67.1317	67.1317	66.5232	66.5232	66.4471	66.6753	67.2077	67.3598	67.7402	(39)
Average = Sum(39)m / 12 =												67.1253	
HLP	Jan 0.9673	Feb 0.9629	Mar 0.9619	Apr 0.9575	May 0.9575	Jun 0.9488	Jul 0.9488	Aug 0.9478	Sep 0.9510	Oct 0.9586	Nov 0.9608	Dec 0.9662	(40)
HLP (average)												0.9574	
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31	

4. Water heating energy requirements (kWh/year)

Assumed occupancy	2.2488 (42)											
Hot water usage for mixer showers	79.5240	78.3289	76.5874	73.2554	70.7965	68.0543	66.4956	68.2239	70.1185	73.0627	76.4663	79.2193
Hot water usage for baths	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Hot water usage for other uses	37.6599	36.2904	34.9210	33.5515	32.1821	30.8126	30.8126	32.1821	33.5515	34.9210	36.2904	37.6599
Average daily hot water use (litres/day)												107.5547
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Energy conte	117.1839	114.6193	111.5084	106.8070	102.9786	98.8669	97.3083	100.4060	103.6700	107.9837	112.7568	116.8792
Energy content (annual)	185.5906	163.2164	171.3843	146.2134	138.6289	121.5684	117.6859	124.3236	127.8355	146.5329	160.6427	183.0046
Distribution loss (46)m = 0.15 x (45)m	27.8386	24.4825	25.7076	21.9320	20.7943	18.2353	17.6529	18.6485	19.1753	21.9799	24.0964	27.4507
Water storage loss:												
Store volume												150.0000
a) If manufacturer declared loss factor is known (kWh/day):												1.9900

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Temperature factor from Table 2b												0.5400 (49)
Enter (49) or (54) in (55)												1.0746 (55)
Total storage loss	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (56)
If cylinder contains dedicated solar storage	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126 (57)
Primary loss	23.2624	21.0112	23.2624	22.5120	23.2624	22.5120	23.2624	23.2624	22.5120	23.2624	22.5120	23.2624 (59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
Total heat required for water heating calculated for each month	242.1656	214.3164	227.9593	200.9634	195.2039	176.3184	174.2609	180.8986	182.5855	203.1079	215.3927	239.5796 (62)
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)
Solar input	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)
Output from w/h	242.1656	214.3164	227.9593	200.9634	195.2039	176.3184	174.2609	180.8986	182.5855	203.1079	215.3927	239.5796 (64)
Total per year (kWh/year) = Sum(64)m =												2452.7522 (64)
Electric shower(s)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =												0.0000 (64a)
Heat gains from water heating, kWh/month	106.9689	95.1495	102.2453	92.4159	91.3541	84.2215	84.3906	86.5976	86.3053	93.9822	97.2137	106.1090 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
(66)m	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	(66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	26.2582	23.3223	18.9670	14.3592	10.7337	9.0618	9.7916	12.7275	17.0828	21.6906	25.3161	26.9880	(67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	294.8562	297.9159	290.2056	273.7913	253.0711	233.5971	220.5873	217.5276	225.2379	241.6522	262.3724	281.8464	(68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	(69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(70)
Losses e.g. evaporation (negative values) (Table 5)	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	(71)
Water heating gains (Table 5)	143.7754	141.5915	137.4264	128.3555	122.7878	116.9743	113.4282	116.3946	119.8685	126.3201	135.0190	142.6197	(72)
Total internal gains	560.6072	558.5470	542.3164	512.2233	482.3100	455.3506	439.5245	442.3671	457.9066	485.3804	518.4249	547.1714	(73)

6. Solar gains

[Jan]														
	Area				Solar flux		g		FF		Access		Gains	
	m2				Table 6a		Specific data		Specific data		factor		W	
					W/m2		or Table 6b		or Table 6c		Table 6d			
North	0.8500				14.5262		0.7600		0.7000		0.7700		4.5521 (74)	
East	1.8100				27.2276		0.7600		0.7000		0.7700		18.1691 (76)	
South	3.3600				60.1191		0.7600		0.7000		0.7700		74.4726 (78)	
West	14.5600				27.2276		0.7600		0.7000		0.7700		146.1556 (80)	
Solar gains	243.3494	388.6301	591.6630	833.2747	942.9850	1044.4253	904.9906	853.7211	705.3102	468.9564	286.3228	197.2775	(83)	
Total gains	803.9566	947.1771	1133.9794	1345.4981	1425.2950	1499.7759	1344.5151	1296.0882	1163.2168	954.3367	804.7477	744.4489	(84)	

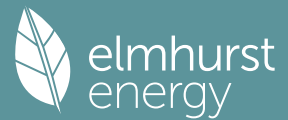
7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	41.1244	41.3097	41.3563	41.5437	41.5437	41.9237	41.9237	41.9717	41.8281	41.4967	41.4030	41.1706
alpha	3.7416	3.7540	3.7571	3.7696	3.7696	3.7949	3.7949	3.7981	3.7885	3.7664	3.7602	3.7447
util living area												
	0.8421	0.7742	0.6642	0.5255	0.4013	0.2735	0.2221	0.2250	0.3399	0.5497	0.7523	0.8549 (86)
Living	20.4580	20.5927	20.7418	20.8485	20.8960	20.9146	20.9173	20.9173	20.9101	20.8574	20.6827	20.4476
Non living	19.5121	19.6687	19.8340	19.9490	19.9957	20.0192	20.0212	20.0222	20.0144	19.9622	19.7790	19.5030
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.7227	20.5927	20.7418	20.8485	20.8960	20.9146	20.9173	20.9173	20.9101	20.8574	20.6827	20.5248 (87)
Th 2	20.1107	20.1143	20.1152	20.1189	20.1189	20.1262	20.1262	20.1271	20.1244	20.1180	20.1162	20.1116 (88)
util rest of house												
	0.8186	0.7458	0.6305	0.4893	0.3628	0.2355	0.1792	0.1806	0.2917	0.4997	0.7154	0.8316 (89)
MIT 2	19.8720	19.6687	19.8340	19.9490	19.9957	20.0192	20.0212	20.0222	20.0144	19.9622	19.7790	19.6126 (90)
Living area fraction									fLA =	Living area / (4) =		
MIT	20.1687	19.9909	20.1506	20.2627	20.3097	20.3315	20.3337	20.3344	20.3268	20.2744	20.0941	19.9308 (91)
Temperature adjustment												0.0000
adjusted MIT	20.1687	19.9909	20.1506	20.2627	20.3097	20.3315	20.3337	20.3344	20.3268	20.2744	20.0941	19.9308 (93)

8. Space heating requirement

Utilisation	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Useful gains	659.4048	702.3748	716.5272	666.4582	529.1314	366.6275	254.6533	247.7613	352.3984	485.6153	574.9423	615.1746	(94)
Ext temp.	7.3000	7.5000	8.4000	9.9000	12.3000	14.8000	16.5000	16.6000	15.0000	12.7000	10.1000	7.8000	(95)
Heat loss rate W	872.7068	843.2886	792.4094	695.6656	537.7044	367.9714	255.0297	248.1378	355.1635	509.0566	673.2028	821.7394	(97)
Space heating kWh	158.6967	94.6941	56.4563	21.0293	6.3783	0.0000	0.0000	0.0000	0.0000	17.4403	70.7475	153.6842	(98a)
Space heating requirement - total per year (kWh/year)												579.1269	
Solar heating kWh	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(98b)
Solar heating contribution - total per year (kWh/year)												0.0000	
Space heating kWh	158.6967	94.6941	56.4563	21.0293	6.3783	0.0000	0.0000	0.0000	0.0000	17.4403	70.7475	153.6842	(98c)

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Space heating requirement after solar contribution - total per year (kWh/year)
Space heating per m2

(98c) / (4) = 579.1269
8.2603 (99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)
Fraction of space heat from main system(s)												1.0000 (202)
Efficiency of main space heating system 1 (in %)												401.5604 (206)
Efficiency of main space heating system 2 (in %)												0.0000 (207)
Efficiency of secondary/supplementary heating system, %												0.0000 (208)
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Space heating requirement	158.6967	94.6941	56.4563	21.0293	6.3783	0.0000	0.0000	0.0000	0.0000	17.4403	70.7475	153.6842 (98)
Space heating efficiency (main heating system 1)	401.5604	401.5604	401.5604	401.5604	401.5604	0.0000	0.0000	0.0000	0.0000	401.5604	401.5604	401.5604 (210)
Space heating fuel (main heating system)	39.5200	23.5815	14.0592	5.2369	1.5884	0.0000	0.0000	0.0000	0.0000	4.3431	17.6182	38.2718 (211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating												
Water heating requirement	242.1656	214.3164	227.9593	200.9634	195.2039	176.3184	174.2609	180.8986	182.5855	203.1079	215.3927	239.5796 (64)
Efficiency of water heater	204.5593	204.5593	204.5593	204.5593	204.5593	204.5593	204.5593	204.5593	204.5593	204.5593	204.5593	204.5593 (216)
Fuel for water heating, kWh/month	118.3841	104.7698	111.4392	98.2421	95.4266	86.1943	85.1885	88.4334	89.2580	99.2905	105.2960	117.1199 (219)
Space cooling fuel requirement	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (221)
Pumps and Fa	11.5590	10.4404	11.5590	11.1861	11.5590	11.1861	11.5590	11.1861	11.5590	11.1861	11.5590	11.5590 (231)
Lighting	22.9836	18.4383	16.6017	12.1631	9.3951	7.6759	8.5705	11.1403	14.4702	18.9857	21.4443	23.6224 (232)
Electricity generated by PVs (Appendix M) (negative quantity)	-51.3923	-70.5109	-108.9662	-127.5054	-135.2224	-129.5994	-123.8020	-120.2480	-105.9558	-84.8206	-55.7797	-42.1498 (233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235c)
Electricity generated by PVs (Appendix M) (negative quantity)	-27.8884	-57.2229	-137.4466	-247.8162	-331.8906	-382.5798	-330.4546	-289.0115	-192.6426	-95.1679	-37.5323	-19.9890 (233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235d)
Annual totals kWh/year												
Space heating fuel - main system 1												144.2191 (211)
Space heating fuel - main system 2												0.0000 (213)
Space heating fuel - secondary												0.0000 (215)
Efficiency of water heater												204.5593
Water heating fuel used												1199.0424 (219)
Space cooling fuel												0.0000 (221)
Electricity for pumps and fans:												
(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6050)												
mechanical ventilation fans (SFP = 0.6050)												136.0977 (230a)
Total electricity for the above, kWh/year												136.0977 (231)
Electricity for lighting (calculated in Appendix L)												185.4911 (232)
Energy saving/generation technologies (Appendices M ,N and Q)												
PV generation												-3305.5949 (233)
Wind generation												0.0000 (234)
Hydro-electric generation (Appendix N)												0.0000 (235a)
Electricity generated - Micro CHP (Appendix N)												0.0000 (235)
Appendix Q - special features												
Energy saved or generated												-0.0000 (236)
Energy used												0.0000 (237)
Total delivered energy for all uses												-1640.7445 (238)

10a. Fuel costs - using BEDF prices (572)

	Fuel kWh/year	Fuel price p/kWh	Fuel cost £/year
Space heating - main system 1	144.2191	26.0600	37.5835 (240)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	1199.0424	26.0600	312.4704 (247)
Energy for instantaneous electric shower(s)	0.0000	26.0600	0.0000 (247a)
Pumps, fans and electric keep-hot	136.0977	26.0600	35.4671 (249)
Energy for lighting	185.4911	26.0600	48.3390 (250)
Additional standing charges			0.0000 (251)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1155.9525	26.0600	-301.2412
PV Unit electricity exported	-2149.6423	5.8100	-124.8942
Total			-426.1355 (252)
Total energy cost			7.7246 (255)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year
Space heating - main system 1	144.2191	0.1582	22.8137 (261)
Total CO2 associated with community systems			0.0000 (373)
Water heating (other fuel)	1199.0424	0.1408	168.8419 (264)
Space and water heating			191.6556 (265)

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Pumps, fans and electric keep-hot	136.0977	0.1387	18.8784 (267)
Energy for lighting	185.4911	0.1443	26.7721 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1155.9525	0.1340	-154.9302
PV Unit electricity exported	-2149.6423	0.1226	-263.5594
Total			-418.4896 (269)
Total CO2, kg/year			-181.1835 (272)

13a. Primary energy - Individual heating systems including micro-CHP

	Energy kWh/year	Primary energy factor kg CO2/kWh	Primary energy kWh/year
Space heating - main system 1	144.2191	1.5855	228.6648 (275)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)			1823.3535 (278)
Space and water heating	1199.0424	1.5207	2052.0183 (279)
Pumps, fans and electric keep-hot	136.0977	1.5128	205.8887 (281)
Energy for lighting	185.4911	1.5338	284.5125 (282)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1155.9525	1.4953	-1728.5095
PV Unit electricity exported	-2149.6423	0.4499	-967.1906
Total			-2695.7001 (283)
Total Primary energy kWh/year			-153.2808 (286)

SAP 10 EPC IMPROVEMENTS

Land Adj to Piskey Laden

Current energy efficiency rating:	A 98
Current environmental impact rating:	A 101

	SAP change	Cost change	CO2 change
N Solar water heating	+ 1.7	-£ 69	-41 kg (22.6%)
U Solar photovoltaic panels			
V2 Wind turbine	+ 24.0	-£ 715	-496 kg (223.3%)

Recommended measures	Typical annual savings	Energy efficiency	Environmental impact
Solar water heating	£69	0.58 kg/m²	A 100 A 101
Wind turbine	£715	7.07 kg/m²	A 124 A 107
Total Savings	£783	7.66 kg/m²	

Potential energy efficiency rating:	A 124
Potential environmental impact rating:	A 107

Fuel prices for cost data on this page from database revision number 572 TEST (30 May 2025)
Recommendation texts revision number 6.1 (11 Jun 2019)

Typical heating and lighting costs of this home (per year, South West England):

	Current £434	Potential £348	Saving £86
Electricity			
Space heating	£73	£94	-£21
Water heating	£312	£206	£107
Lighting	£48	£48	£0
Generated (PV)	-£426	-£409	-£17
Generated (wind)	-£0	-£715	£715
Total cost of fuels	£8	-£776	£784
Total cost of uses	£7	-£776	£784
Delivered energy	-23 kWh/m²	-79 kWh/m²	56 kWh/m²
Carbon dioxide emissions	-0.2 tonnes	-0.7 tonnes	0.5 tonnes
CO2 emissions per m²	-3 kg/m²	-10 kg/m²	8 kg/m²
Primary energy	-2 kWh/m²	-70 kWh/m²	67 kWh/m²

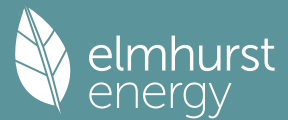
SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022)
CALCULATION OF ENERGY RATING FOR IMPROVED DWELLING

1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)
Main dwelling	70.1100 (1b)	x 2.6300 (2b)	= 184.3893 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100		= 184.3893 (4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n)	= 184.3893 (5)

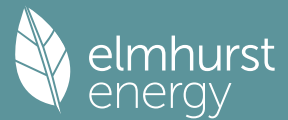
2. Ventilation rate

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												m3 per hour	
Number of open chimneys												0 * 80 = 0.0000 (6a)	
Number of open flues												0 * 20 = 0.0000 (6b)	
Number of chimneys / flues attached to closed fire												0 * 10 = 0.0000 (6c)	
Number of flues attached to solid fuel boiler												0 * 20 = 0.0000 (6d)	
Number of flues attached to other heater												0 * 35 = 0.0000 (6e)	
Number of blocked chimneys												0 * 20 = 0.0000 (6f)	
Number of intermittent extract fans												0 * 10 = 0.0000 (7a)	
Number of passive vents												0 * 10 = 0.0000 (7b)	
Number of flueless gas fires												0 * 40 = 0.0000 (7c)	
												Air changes per hour	
Infiltration due to chimneys, flues and fans	= (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =											0.0000 / (5) = 0.0000 (8)	
Pressure test												Yes	
Pressure Test Method												Blower Door	
Measured/design AP50												1.0000 (17)	
Infiltration rate												0.0500 (18)	
Number of sides sheltered												0 (19)	
Shelter factor	(20) = 1 - [0.075 x (19)] =											1.0000 (20)	
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) =											0.0500 (21)	
Wind speed	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Wind factor	5.1000	5.0000	4.9000	4.4000	4.3000	3.8000	3.8000	3.7000	4.0000	4.3000	4.5000	4.7000	(22)
Adj infilt rate	1.2750	1.2500	1.2250	1.1000	1.0750	0.9500	0.9500	0.9250	1.0000	1.0750	1.1250	1.1750	(22a)
Balanced mechanical ventilation with heat recovery	0.0638	0.0625	0.0613	0.0550	0.0537	0.0475	0.0475	0.0463	0.0500	0.0537	0.0563	0.0588	(22b)
If mechanical ventilation													0.5000 (23a)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)													0.5000 (23b)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =													84.6000 (23c)
Effective ac	0.1407	0.1395	0.1382	0.1320	0.1307	0.1245	0.1245	0.1232	0.1270	0.1307	0.1332	0.1357	(25)
3. Heat losses and heat loss parameter													
Element	Gross	Openings	NetArea	U-value	A x U	K-value	A x K						
Main dwelling	m2	m2	m2	W/m2K	W/K	kJ/m2K	kJ/K						
Door			2.1200	1.0000	2.1200								
Window			20.5800	1.1450	23.5649								
Heatloss Floor 1			70.1100	0.1200	8.4132	110.0000	7712.1000						
External Wall 1	81.6400	22.7000	58.9400	0.1600	9.4304	9.0000	530.4600						
External Roof 1	72.5800		72.5800	0.1400	10.1612	9.0000	653.2200						
Total net area of external elements Aum(A, m2)			224.3300										
Fabric heat loss, W/K = Sum (A x U)			(26)...(30) + (32) =		53.6897								
Main dwelling													(33)
Internal Wall 1			127.1400					9.0000	1144.2600 (32c)				
Heat capacity Cm = Sum(A x k)													(28)...(30) + (32) + (32a)...(32e) = 10040.0400 (34)
Thermal mass parameter (TMP = Cm / TFA) in kJ/m2K													143.2041 (35)
List of Thermal Bridges													
K1 Element					Length	Psi-value	Total						
E2 Other lintels (including other steel lintels)					13.0700	0.0840	1.0979						
E4 Jamb					32.8000	0.0430	1.4104						
E5 Ground floor (normal)					35.9600	0.0210	0.7552						
E3 Sill					4.9700	0.0340	0.1690						
E11 Eaves (insulation at rafter level)					24.2600	0.0400	0.9704						
E13 Gable (insulation at rafter level)					11.8000	0.0240	0.2832						
E16 Corner (normal)					8.9200	0.0300	0.2676						
Thermal bridges (Sum(L x Psi) calculated using Appendix K)							4.9536	(36) (36)					
Point Thermal bridges							0.0000						
Total fabric heat loss							(33) + (36) + (36a) =	58.6433 (37)					
Ventilation heat loss calculated monthly (38)m = 0.33 x (25)m x (5)													
(38)m	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Heat transfer coeff	8.5644	8.4884	8.4123	8.0320	7.9559	7.5756	7.5756	7.4996	7.7278	7.9559	8.1081	8.2602	(38)
Average = Sum(39)m / 12 =	67.2077	67.1317	67.0556	66.6753	66.5992	66.2189	66.2189	66.1429	66.3711	66.5992	66.7514	66.9035	(39)
													66.6563
HLP	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
HLP (average)	0.9586	0.9575	0.9564	0.9510	0.9499	0.9445	0.9445	0.9434	0.9467	0.9499	0.9521	0.9543	(40)
Days in mont	31	28	31	30	31	30	31	31	30	31	30	31	
													0.9507
4. Water heating energy requirements (kWh/year)													
Assumed occupancy													2.2488 (42)
Hot water usage for mixer showers													
Hot water usage for baths	79.5240	78.3289	76.5874	73.2554	70.7965	68.0543	66.4956	68.2239	70.1185	73.0627	76.4663	79.2193	(42a)
Hot water usage for other uses	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	(42b)
Average daily hot water use (litres/day)	37.6599	36.2904	34.9210	33.5515	32.1821	30.8126	30.8126	32.1821	33.5515	34.9210	36.2904	37.6599	(42c)
													107.5547 (43)
Daily hot water use	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Energy conte	117.1839	114.6193	111.5084	106.8070	102.9786	98.8669	97.3083	100.4060	103.6700	107.9837	112.7568	116.8792	(44)
Energy content (annual)	185.5906	163.2164	171.3843	146.2134	138.6289	121.5684	117.6859	124.3236	127.8355	146.5329	160.6427	183.0046	(45)
Distribution loss (46)m = 0.15 x (45)m	27.8386	24.4825	25.7076	21.9320	20.7943	18.2353	17.6529	18.6485	19.1753	21.9799	24.0964	27.4507	(46)
Water storage loss:													
Store volume													150.0000 (47)
a) If manufacturer declared loss factor is known (kWh/day):													1.9900 (48)
Temperature factor from Table 2b													0.5400 (49)
Enter (49) or (54) in (55)													1.0746 (55)
Total storage loss	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126	(56)
If cylinder contains dedicated solar storage	33.3126	30.0888	33.3126	32.2380	33.3126	32.2380	33.3126	33.3126	32.2380	33.3126	32.2380	33.3126	(57)

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Primary loss	23.2624	21.0112	21.8667	15.7584	10.4681	9.9053	10.2355	11.1660	17.1091	21.8667	22.5120	23.2624 (59)
Combi loss	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (61)
Total heat required for water heating calculated for each month												
	242.1656	214.3164	226.5635	194.2098	182.4096	163.7117	161.2340	168.8022	177.1826	201.7122	215.3927	239.5796 (62)
WWHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63a)
PV diverter	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (63b)
Aperture area of solar collector												3.0000 (H1)
Zero-loss collector efficiency												0.8000 (H2)
Collector linear heat loss coefficient												1.8000 (H3)
Collector 2nd order heat loss coefficient												0.0000 (H4)
Collector loop efficiency												0.9000 (H5)
Incidence angle modifier												1.0000 (H6)
Overshading factor												0.8000 (H8)
Overall heat loss coefficient of system												6.5000 (H10)
Heat loss coefficient of collector loop												3.9667 (H11)
Dedicated solar storage volume												75.0000 (H12)
Effective solar volume												75.0000 (H14)
Reference volume												225.0000 (H15)
Storage tank correction coefficient												1.3161 (H16)
Heat delivered to hot water												606.3511 (H24)
Heat delivered to space heating												0.0000 (H29)
Solar input												606.3511
Solar input	-0.0000	-16.2364	-57.5166	-78.3835	-101.3807	-93.3491	-92.6415	-81.4867	-56.7929	-28.5636	-0.0000	-0.0000 (63c)
FGHRS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (63d)
Output from w/h	242.1656	198.0800	169.0470	115.8263	81.0289	70.3626	68.5924	87.3155	120.3897	173.1486	215.3927	239.5796 (64)
Total per year (kWh/year) = Sum(64)m =												1780.9288 (64)
Electric shower(s)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (64a)
Total Energy used by instantaneous electric shower(s) (kWh/year) = Sum(64a)m =												0.0000 (64a)
Heat gains from water heating, kWh/month	106.9689	95.1495	101.1287	87.0131	81.1187	74.1361	73.9690	76.9204	81.9830	92.8656	97.2137	106.1090 (65)

5. Internal gains (see Table 5 and 5a)

Metabolic gains (Table 5), Watts												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
(66)m	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275	134.9275 (66)
Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	26.2582	23.3223	18.9670	14.3592	10.7337	9.0618	9.7916	12.7275	17.0828	21.6906	25.3161	26.9880 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	294.8562	297.9159	290.2056	273.7913	253.0711	233.5971	220.5873	217.5276	225.2379	241.6522	262.3724	281.8464 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415 (69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517 (71)
Water heating gains (Table 5)	143.7754	141.5915	135.9256	120.8515	109.0305	102.9668	99.4207	103.3877	113.8653	124.8193	135.0190	142.6197 (72)
Total internal gains	560.6072	558.5470	540.8156	504.7193	468.5527	441.3432	425.5170	429.3602	451.9034	483.8796	518.4249	547.1714 (73)

6. Solar gains

[Jan]				Area m2	Solar flux Table 6a W/m2		g Specific data or Table 6b		FF Specific data or Table 6c		Access factor Table 6d		Gains W
North				0.8500	10.6334		0.7600		0.7000		0.7700		3.3322 (74)
East				1.8100	19.6403		0.7600		0.7000		0.7700		13.1060 (76)
South				3.3600	46.7521		0.7600		0.7000		0.7700		57.9142 (78)
West				14.5600	19.6403		0.7600		0.7000		0.7700		105.4274 (80)
Solar gains	179.7799	333.0931	513.5087	710.8654	848.2508	860.7078	822.3937	719.8947	583.3490	385.0244	220.5561	150.2972	(83)
Total gains	740.3870	891.6402	1054.3243	1215.5847	1316.8034	1302.0510	1247.9107	1149.2549	1035.2524	868.9040	738.9810	697.4686	(84)

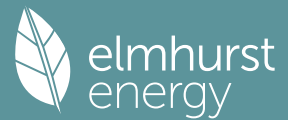
7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)												21.0000 (85)
Utilisation factor for gains for living area, nil,m (see Table 9a)												
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	41.4967	41.5437	41.5909	41.8281	41.8759	42.1163	42.1163	42.1648	42.0198	41.8759	41.7804	41.6854
alpha	3.7664	3.7696	3.7727	3.7885	3.7917	3.8078	3.8078	3.8110	3.8013	3.7917	3.7854	3.7790
util living area	0.9176	0.8589	0.7573	0.6087	0.4557	0.3224	0.2328	0.2635	0.4310	0.6938	0.8704	0.9304 (86)
Living	20.1085	20.3423	20.6004	20.7962	20.8833	20.9116	20.9174	20.9165	20.8973	20.7542	20.3973	20.0490
Non living	19.0919	19.3752	19.6787	19.9006	19.9906	20.0209	20.0251	20.0256	20.0082	19.8640	19.4521	19.0219
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.5440	20.3423	20.6004	20.7962	20.8833	20.9116	20.9174	20.9165	20.8973	20.7542	20.3973	20.1820 (87)
Th 2	20.1180	20.1189	20.1198	20.1244	20.1253	20.1298	20.1298	20.1308	20.1280	20.1253	20.1234	20.1216 (88)
util rest of house	0.9054	0.8405	0.7304	0.5739	0.4163	0.2796	0.1870	0.2142	0.3800	0.6533	0.8500	0.9197 (89)
MIT 2	19.7089	19.3752	19.6787	19.9006	19.9906	20.0209	20.0251	20.0256	20.0082	19.8640	19.4521	19.2200 (90)
Living area fraction									fLA = Living area / (4) =			0.3487 (91)
MIT	20.0001	19.7124	20.0001	20.2129	20.3019	20.3316	20.3363	20.3363	20.3182	20.1744	19.7817	19.5555 (92)
Temperature adjustment												0.0000
adjusted MIT	20.0001	19.7124	20.0001	20.2129	20.3019	20.3316	20.3363	20.3363	20.3182	20.1744	19.7817	19.5555 (93)

8. Space heating requirement

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Utilisation	0.9036	0.8311	0.7265	0.5775	0.4244	0.2896	0.1979	0.2259	0.3913	0.6557	0.8411	0.9118 (94)
Useful gains	669.0450	741.0601	765.9399	702.0308	558.8296	377.0701	246.9942	259.6530	405.0940	569.7368	621.5583	635.9798 (95)
Ext temp.	4.3000	4.9000	6.5000	8.9000	11.7000	14.6000	16.6000	16.4000	14.1000	10.6000	7.1000	4.2000 (96)

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Heat loss rate W	1055.1691	994.3835	905.2594	754.2925	572.8787	379.5372	247.4127	260.3558	412.7113	637.6508	846.5225	1027.3375 (97)
Space heating kWh	287.2764	170.2333	103.6537	37.6284	10.4525	0.0000	0.0000	0.0000	0.0000	50.5281	161.9742	291.1701 (98a)
Space heating requirement - total per year (kWh/year)												1112.9167
Solar heating kWh	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (98b)
Solar heating contribution - total per year (kWh/year)												0.0000
Space heating kWh	287.2764	170.2333	103.6537	37.6284	10.4525	0.0000	0.0000	0.0000	0.0000	50.5281	161.9742	291.1701 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)												1112.9167
Space heating per m2										(98c) / (4) =		15.8739 (99)

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)												0.0000 (201)
Fraction of space heat from main system(s)												1.0000 (202)
Efficiency of main space heating system 1 (in %)												401.8421 (206)
Efficiency of main space heating system 2 (in %)												0.0000 (207)
Efficiency of secondary/supplementary heating system, %												0.0000 (208)
Space heating requirement	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
287.2764	170.2333	103.6537	37.6284	10.4525	0.0000	0.0000	0.0000	0.0000	0.0000	50.5281	161.9742	291.1701 (98)
Space heating efficiency (main heating system 1)	401.8421	401.8421	401.8421	401.8421	401.8421	0.0000	0.0000	0.0000	0.0000	401.8421	401.8421	401.8421 (210)
Space heating fuel (main heating system)	71.4899	42.3632	25.7946	9.3640	2.6011	0.0000	0.0000	0.0000	0.0000	12.5741	40.3079	72.4588 (211)
Space heating efficiency (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating												
Water heating requirement	242.1656	198.0800	169.0470	115.8263	81.0289	70.3626	68.5924	87.3155	120.3897	173.1486	215.3927	239.5796 (64)
Efficiency of water heater	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890	204.5890 (216)
Fuel for water heating, kWh/month	118.3669	96.8185	82.6276	56.6141	39.6057	34.3922	33.5269	42.6785	58.8446	84.6324	105.2807	117.1028 (219)
Space cooling fuel requirement												
(221)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (221)
Pumps and Fa	18.3535	16.5774	18.3535	17.7615	18.3535	17.7615	18.3535	18.3535	17.7615	18.3535	17.7615	18.3535 (231)
Lighting	22.9836	18.4383	16.6017	12.1631	9.3951	7.6759	8.5705	11.1403	14.4702	18.9857	21.4443	23.6224 (232)
Electricity generated by PVs (Appendix M) (negative quantity)												
(233a)m	-39.4027	-61.8717	-96.7434	-111.0759	-116.5537	-107.9993	-106.3212	-102.0549	-90.9012	-72.7467	-44.8718	-32.8796 (233a)
Electricity generated by wind turbines (Appendix M) (negative quantity)												
(234a)m	-212.5732	-192.0016	-212.5732	-205.7160	-212.5732	-205.7160	-212.5732	-212.5732	-205.7160	-212.5732	-205.7160	-212.5732 (234a)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												
(235a)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235a)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)												
(235c)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235c)
Electricity generated by PVs (Appendix M) (negative quantity)												
(233b)m	-16.3080	-42.6377	-108.9577	-199.9641	-294.8486	-306.7207	-298.6529	-234.3521	-147.5628	-68.6728	-23.5570	-12.1175 (233b)
Electricity generated by wind turbines (Appendix M) (negative quantity)												
(234b)m	-91.1028	-82.2864	-91.1028	-88.1640	-91.1028	-88.1640	-91.1028	-91.1028	-88.1640	-91.1028	-88.1640	-91.1028 (234b)
Electricity generated by hydro-electric generators (Appendix M) (negative quantity)												
(235b)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235b)
Electricity used or net electricity generated by micro-CHP (Appendix N) (negative if net generation)												
(235d)m	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (235d)
Annual totals kWh/year												
Space heating fuel - main system 1												276.9537 (211)
Space heating fuel - main system 2												0.0000 (213)
Space heating fuel - secondary												0.0000 (215)
Efficiency of water heater												204.5890
Water heating fuel used												870.4908 (219)
Space cooling fuel												0.0000 (221)

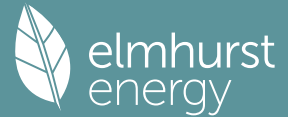
Electricity for pumps and fans:												
(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6050)												
mechanical ventilation fans (SFP = 0.6050)												136.0977 (230a)
pump for solar water heating												80.0000 (230g)
Total electricity for the above, kWh/year												216.0977 (231)
Electricity for lighting (calculated in Appendix L)												185.4911 (232)

Energy saving/generation technologies (Appendices M ,N and Q)												
PV generation												-2737.7741 (233)
Wind generation												-3575.5408 (234)
Hydro-electric generation (Appendix N)												0.0000 (235a)
Electricity generated - Micro CHP (Appendix N)												0.0000 (235)
Appendix Q - special features												
Energy saved or generated												-0.0000 (236)
Energy used												0.0000 (237)
Total delivered energy for all uses												-4764.2814 (238)

10a. Fuel costs - using Table 12 prices

	Fuel kWh/year	Fuel price p/kWh	Fuel cost £/year
Space heating - main system 1	276.9537	16.4900	45.6697 (240)
Total CO2 associated with community systems			0.0000 (473)
Water heating (other fuel)	870.4908	16.4900	143.5439 (247)
Energy for instantaneous electric shower(s)	0.0000	16.4900	0.0000 (247a)
Pumps, fans and electric keep-hot	136.0977	16.4900	22.4425 (249)
Pump for solar water heating	80.0000	16.4900	13.1920 (249)
Energy for lighting	185.4911	16.4900	30.5875 (250)
Additional standing charges			0.0000 (251)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-983.4220	16.4900	-162.1663
PV Unit electricity exported	-1754.3521	5.5900	-98.0683
Total			-260.2346 (252)
Wind Turbine electricity used in dwelling	-2502.8785	16.4900	-412.7247
Wind Turbine electricity exported	-1072.6622	5.5900	-59.9618

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Total -472.6865 (252)
Total energy cost -477.4854 (255)

11a. SAP rating - Individual heating systems

Energy cost deflator (Table 12): 0.3600 (256)
Energy cost factor (ECF) [(255) x (256)] / [(4) + 45.0] = -1.4933 (257)
SAP value 124.2065
SAP rating (Section 12) 124 (258)
SAP band A

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP

	Energy kWh/year	Emission factor kg CO2/kWh	Emissions kg CO2/year
Space heating - main system 1	276.9537	0.1577	43.6673 (261)
Total CO2 associated with community systems			0.0000 (373)
Water heating (other fuel)	870.4908	0.1460	127.0533 (264)
Space and water heating			170.7206 (265)
Pumps, fans and electric keep-hot	216.0977	0.1387	29.9754 (267)
Energy for lighting	185.4911	0.1443	26.7721 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-983.4220	0.1339	-131.7030
PV Unit electricity exported	-1754.3521	0.1213	-212.8554
Total			-344.5584 (269)
Wind Turbine electricity used in dwelling	-2502.8785	0.1387	-347.1801
Wind Turbine electricity exported	-1072.6622	0.1387	-148.7915
Total			-495.9716 (269)
Total CO2, kg/year			-613.0619 (272)
CO2 emissions per m2			-8.7400 (273)
EI value			107.1367
EI rating			107 (274)
EI band			A

SAP 10 WORKSHEET FOR New Build (As Designed) (Version 10.2, February 2022) CALCULATION OF EPC COSTS, EMISSIONS AND PRIMARY ENERGY FOR IMPROVED DWELLING

1. Overall dwelling characteristics

	Area (m2)	Storey height (m)	Volume (m3)
Main dwelling	70.1100 (1b)	2.6300 (2b)	184.3893 (1b) - (3b)
Ground floor			
Total floor area TFA = (1a)+(1b)+(1c)+(1d)+(1e)...(1n)	70.1100		(4)
Dwelling volume		(3a)+(3b)+(3c)+(3d)+(3e)...(3n)	184.3893 (5)

2. Ventilation rate

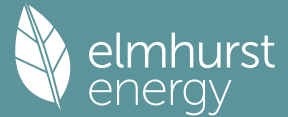
	m3 per hour
Number of open chimneys	0 * 80 = 0.0000 (6a)
Number of open flues	0 * 20 = 0.0000 (6b)
Number of chimneys / flues attached to closed fire	0 * 10 = 0.0000 (6c)
Number of flues attached to solid fuel boiler	0 * 20 = 0.0000 (6d)
Number of flues attached to other heater	0 * 35 = 0.0000 (6e)
Number of blocked chimneys	0 * 20 = 0.0000 (6f)
Number of intermittent extract fans	0 * 10 = 0.0000 (7a)
Number of passive vents	0 * 10 = 0.0000 (7b)
Number of flueless gas fires	0 * 40 = 0.0000 (7c)
Infiltration due to chimneys, flues and fans = (6a)+(6b)+(6c)+(6d)+(6e)+(6f)+(6g)+(7a)+(7b)+(7c) =	0.0000 / (5) = 0.0000 (8)
Pressure test	Yes
Pressure Test Method	Blower Door
Measured/design AP50	1.0000 (17)
Infiltration rate	0.0500 (18)
Number of sides sheltered	0 (19)
Shelter factor	(20) = 1 - [0.075 x (19)] = 1.0000 (20)
Infiltration rate adjusted to include shelter factor	(21) = (18) x (20) = 0.0500 (21)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind speed	5.9000	5.5000	5.4000	5.0000	5.0000	4.2000	4.2000	4.1000	4.4000	5.1000	5.3000	5.8000 (22)
Wind factor	1.4750	1.3750	1.3500	1.2500	1.2500	1.0500	1.0500	1.0250	1.1000	1.2750	1.3250	1.4500 (22a)
Adj infilt rate	0.0738	0.0688	0.0675	0.0625	0.0625	0.0525	0.0525	0.0512	0.0550	0.0638	0.0663	0.0725 (22b)
Balanced mechanical ventilation with heat recovery												0.5000 (23a)
If mechanical ventilation												0.5000 (23b)
If exhaust air heat pump using Appendix N, (23b) = (23a) x Fmv (equation (N5)), otherwise (23b) = (23a)												84.6000 (23c)
If balanced with heat recovery: efficiency in % allowing for in-use factor (from Table 4h) =												
Effective ac	0.1507	0.1457	0.1445	0.1395	0.1395	0.1295	0.1295	0.1282	0.1320	0.1407	0.1432	0.1495 (25)

3. Heat losses and heat loss parameter

[illegible]

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Lighting gains (calculated in Appendix L, equation L9 or L9a), also see Table 5	26.2582	23.3223	18.9670	14.3592	10.7337	9.0618	9.7916	12.7275	17.0828	21.6906	25.3161	26.9880 (67)
Appliances gains (calculated in Appendix L, equation L13 or L13a), also see Table 5	294.8562	297.9159	290.2056	273.7913	253.0711	233.5971	220.5873	217.5276	225.2379	241.6522	262.3724	281.8464 (68)
Cooking gains (calculated in Appendix L, equation L15 or L15a), also see Table 5	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415	50.7415 (69)
Pumps, fans	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (70)
Losses e.g. evaporation (negative values) (Table 5)	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517	-89.9517 (71)
Water heating gains (Table 5)	143.7754	141.5915	135.9256	120.8515	109.0305	102.9668	99.4207	103.3877	113.8653	124.8193	135.0190	142.6197 (72)
Total internal gains	560.6072	558.5470	540.8156	504.7193	468.5527	441.3432	425.5170	429.3602	451.9034	483.8796	518.4249	547.1714 (73)

6. Solar gains

[Jan]					Area m2	Solar flux Table 6a W/m2		g Specific data or Table 6b	FF Specific data or Table 6c	Access factor Table 6d		Gains W
North					0.8500	14.5262		0.7600	0.7000	0.7700		4.5521 (74)
East					1.8100	27.2276		0.7600	0.7000	0.7700		18.1691 (76)
South					3.3600	60.1191		0.7600	0.7000	0.7700		74.4726 (78)
West					14.5600	27.2276		0.7600	0.7000	0.7700		146.1556 (80)
Solar gains	243.3494	388.6301	591.6630	833.2747	942.9850	1044.4253	904.9906	853.7211	705.3102	468.9564	286.3228	197.2775 (83)
Total gains	803.9566	947.1771	1132.4786	1337.9941	1411.5377	1485.7684	1330.5076	1283.0812	1157.2136	952.8359	804.7477	744.4489 (84)

7. Mean internal temperature (heating season)

Temperature during heating periods in the living area from Table 9, Th1 (C)	21.0000 (85)											
Utilisation factor for gains for living area, nil,m (see Table 9a)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
tau	41.1244	41.3097	41.3563	41.5437	41.5437	41.9237	41.9237	41.9717	41.8281	41.4967	41.4030	41.1706
alpha	3.7416	3.7540	3.7571	3.7696	3.7696	3.7949	3.7949	3.7981	3.7885	3.7664	3.7602	3.7447
util living area	0.8421	0.7742	0.6648	0.5280	0.4049	0.2760	0.2244	0.2272	0.3416	0.5504	0.7523	0.8549 (86)
Living	20.4580	20.5927	20.7412	20.8474	20.8954	20.9145	20.9172	20.9173	20.9100	20.8571	20.6827	20.4476
Non living	19.5121	19.6687	19.8334	19.9480	19.9951	20.0191	20.0212	20.0222	20.0143	19.9620	19.7790	19.5030
24 / 16	0	0	0	0	0	0	0	0	0	0	0	0
24 / 9	3	0	0	0	0	0	0	0	0	0	0	0
16 / 9	28	0	0	0	0	0	0	0	0	0	0	10
MIT	20.7227	20.5927	20.7412	20.8474	20.8954	20.9145	20.9172	20.9173	20.9100	20.8571	20.6827	20.5248 (87)
Th 2	20.1107	20.1143	20.1152	20.1189	20.1189	20.1262	20.1262	20.1271	20.1244	20.1180	20.1162	20.1116 (88)
util rest of house	0.8186	0.7458	0.6311	0.4917	0.3662	0.2377	0.1811	0.1824	0.2932	0.5004	0.7154	0.8316 (89)
MIT 2	19.8720	19.6687	19.8334	19.9480	19.9951	20.0191	20.0212	20.0222	20.0143	19.9620	19.7790	19.6126 (90)
Living area fraction	fLA = Living area / (4) =											
MIT	20.1687	19.9909	20.1500	20.2616	20.3091	20.3314	20.3337	20.3343	20.3266	20.2741	20.0941	19.9308 (92)
Temperature adjustment	0.0000											
adjusted MIT	20.1687	19.9909	20.1500	20.2616	20.3091	20.3314	20.3337	20.3343	20.3266	20.2741	20.0941	19.9308 (93)

8. Space heating requirement

Utilisation	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Useful gains	0.8202	0.7415	0.6325	0.4977	0.3746	0.2467	0.1914	0.1931	0.3045	0.5095	0.7144	0.8263 (94)
Ext temp.	659.4048	702.3748	716.2506	665.9183	528.8223	366.5761	254.6367	247.7452	352.3425	485.4957	574.9423	615.1746 (95)
Heat loss rate W	7.3000	7.5000	8.4000	9.9000	12.3000	14.8000	16.5000	16.6000	15.0000	12.7000	10.1000	7.8000 (96)
Space heating	872.7068	843.2886	792.3707	695.5928	537.6632	367.9644	255.0273	248.1355	355.1559	509.0403	673.2028	821.7394 (97)
Space heating requirement - total per year (kWh/year)	158.6967	94.6941	56.6334	21.3657	6.5776	0.0000	0.0000	0.0000	0.0000	17.5172	70.7475	153.6842 (98a)
Solar heating kWh	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000 (98b)
Solar heating contribution - total per year (kWh/year)	158.6967	94.6941	56.6334	21.3657	6.5776	0.0000	0.0000	0.0000	0.0000	17.5172	70.7475	153.6842 (98c)
Space heating requirement after solar contribution - total per year (kWh/year)	158.6967	94.6941	56.6334	21.3657	6.5776	0.0000	0.0000	0.0000	0.0000	17.5172	70.7475	153.6842 (98c)
Space heating per m2	(98c) / (4) =											

9a. Energy requirements - Individual heating systems, including micro-CHP

Fraction of space heat from secondary/supplementary system (Table 11)	0.0000 (201)											
Fraction of space heat from main system(s)	1.0000 (202)											
Efficiency of main space heating system 1 (in %)	401.5604 (206)											
Efficiency of main space heating system 2 (in %)	0.0000 (207)											
Efficiency of secondary/supplementary heating system, %	0.0000 (208)											
Space heating requirement	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Space heating efficiency (main heating system 1)	158.6967	94.6941	56.6334	21.3657	6.5776	0.0000	0.0000	0.0000	0.0000	17.5172	70.7475	153.6842 (98)
Space heating fuel (main heating system 1)	401.5604	401.5604	401.5604	401.5604	401.5604	0.0000	0.0000	0.0000	0.0000	401.5604	401.5604	401.5604 (210)
Space heating efficiency (main heating system 2)	39.5200	23.5815	14.1033	5.3207	1.6380	0.0000	0.0000	0.0000	0.0000	4.3623	17.6182	38.2718 (211)
Space heating fuel (main heating system 2)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (212)
Space heating fuel (secondary)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (213)
Space heating requirement	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000 (215)
Water heating	232.2296	186.1312	153.0157	97.5560	68.1774	49.7236	58.7224	69.4148	101.7016	155.4781	202.0599	239.5796 (64)

Electricity for pumps and fans:		
(BalancedWithHeatRecovery, Database: in-use factor = 1.1000, SFP = 0.6050)		
mechanical ventilation fans (SFP = 0.6050)	136.0977	(230a)
pump for solar water heating	80.0000	(230g)
Total electricity for the above, kWh/year	216.0977	(231)
Electricity for lighting (calculated in Appendix L)	185.4911	(232)
Energy saving/generation technologies (Appendices M ,N and Q)		
PV generation	-3305.5949	(233)
Wind generation	-3575.5408	(234)
Hydro-electric generation (Appendix N)	0.0000	(235a)
Electricity generated - Micro CHP (Appendix N)	0.0000	(235)
Appendix Q - special features		
Energy saved or generated	-0.0000	(236)
Energy used	0.0000	(237)
Total delivered energy for all uses	-5546.2203	(238)

12a. Carbon dioxide emissions - Individual heating systems including micro-CHP			
	Energy kWh/year	Emission factor kg CO ₂ /kWh	Emissions kg CO ₂ /year
Space heating - main system 1	144.4158	0.1582	22.8416 (261)
Total CO ₂ associated with community systems			0.0000 (373)
Water heating (other fuel)	788.9107	0.1471	116.0399 (264)
Space and water heating			138.8815 (265)
Pumps, fans and electric keep-hot	216.0977	0.1387	29.9754 (267)
Energy for lighting	185.4911	0.1443	26.7721 (268)
Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1070.5817	0.1348	-144.3483
PV Unit electricity exported	-2235.0132	0.1223	-273.3910
Total			-417.7393 (269)
Wind Turbine electricity used in dwelling	-2502.8785	0.1387	-347.1801
Wind Turbine electricity exported	-1072.6622	0.1387	-148.7915
Total			-495.9716 (269)
Total CO ₂ , kg/year			-718.0818 (272)

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Energy saving/generation technologies			
PV Unit electricity used in dwelling	-1070.5817	1.4983	-1604.0924
PV Unit electricity exported	-2235.0132	0.4489	-1003.2393
Total			-2607.3317 (283)
Wind Turbine electricity used in dwelling	-2502.8785	1.5128	-3786.3546
Wind Turbine electricity exported	-1072.6622	0.5128	-550.0612
Total			-4336.4158 (283)
Total Primary energy kWh/year			-4885.1577 (286)